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MedPredict AI: Predictive Intelligence Model for Detecting Pneumonia

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COMMITTEE REPORT

We certify that we have read this graduation project report as examining committee, examined the student in its content and that in our opinion it is adequate as a project document for B.Sc. in Informatics and Computer Systems.

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Signature

ABSTRACT

Pneumonia remains one of the leading causes of morbidity and mortality worldwide, particularly among young children, the elderly, and individuals with weakened immune systems. Traditional diagnostic methods that rely on manual interpretation of chest X-ray (CXR) images are often slow, subjective, and dependent on expert availability, which can lead to delays in diagnosis and treatment. This project, MedPredict AI: Predictive Intelligence Model for Detecting Pneumonia, aims to develop an intelligent diagnostic system that utilizes Convolutional Neural Network (CNN) model for rapid and accurate pneumonia detection. The model is trained and validated using publicly available CXR datasets to classify images as either Normal or Pneumonia with high accuracy, sensitivity, and specificity. The proposed system is integrated within a user-friendly Android application designed to assist healthcare professionals and patients by providing reliable diagnostic support, particularly in resource-limited areas. By combining AI-based image analysis with accessibility and efficiency, this project seeks to enhance early detection, reduce diagnostic errors, and improve overall healthcare outcomes.

TABLE OF CONTENTS

COMMITTEE REPORT	1
ACKNOWLEDGEMENTS	2
ABSTRACT.....	3
TABLE OF CONTENTS.....	4
TABLE OF FIGURES	7
LIST OF TABLES	8
LIST OF ABBREVIATIONS	9
CHAPTER 1: INTRODUCTION	10
1.1 Background of the Study	10
1.2 Problem Statement	10
1.3 Objectives of the project	11
1.4 Research Questions	11
1.5 Scope of the Study.....	12
1.6 Significance of the Study	12
1.7 Project Contribution	13
1.8 Organization of the Report.....	13
1.9 Conclusion.....	14
CHAPTER 2: LITERATURE REVIEW	15
2.1 BACKGROUND.....	15
2.1.1 Types of Pneumonia	15
2.1.2 Role of Imaging in Pneumonia Diagnosis	19
2.1.3 Artificial Intelligence in Healthcare	19
2.1.4 Importance of Early Detection	20
2.2 History.....	20
2.2.1 Early Approaches: Traditional Image Processing and Machine Learning ...	21
2.2.2 The Deep Learning Revolution	21
2.2.3 Growth of Medical Imaging Datasets (Mid-2010s)	21
2.2.4 Current Trends and Challenges (Present Day)	22
2.3 SIMILAR SYSTEMS	22
2.3.1 CheXNet	22
2.3.2 COVID-Net	23

2.3.3 DeepBreath System	24
2.3.4 Qure.ai qXR System	25
2.3.5 Lunit INSIGHT CXR System	26
2.3.6 Zebra Medical Vision System	27
2.3 Comparative Study	28
2.4 Conclusion.....	29
CHAPTER 3: PROJECT MANAGEMENT	30
3.1 Introduction	30
3.2 Approach	30
3.2.1 Justification for Selecting the Waterfall Model:.....	30
3.2.2 Phases of Waterfall Model	31
3.3 Risk Management.....	32
3.4 Project Plan	33
3.4.1 Plan and Gantt Chart.....	35
3.5 Conclusion.....	36
CHAPTER 4: REQUIERMENTS AND ANALYSIS	37
4.1 Introduction	37
4.2 Feasibility Study.....	37
4.2.1 Operational Feasibility	37
4.2.2 Technical Feasibility.....	37
4.2.3 Economic Feasibility	38
4.3 System Requirements	38
4.3.1 Functional Requirements	38
4.3.1.1 Admin Functional Requirements:	39
4.3.1.2 User Functional Requirements:.....	39
4.3.2 Non-Functional Requirements.....	39
4.3.3 Software Requirements.....	40
4.3.4 Hardware Requirements	41
4.4 Personas.....	41
4.5 System Models	43
4.5.1 Use Case Diagrams.....	43
4.5.2 Sequence Diagram	46
4.5.3 Activity Diagram	49

4.6 Conclusion.....	52
CHAPTER 5: SYSTEM DESIGN	53
5.1 Introduction	53
5.2 Product Features	53
5.3 USER INTERFACE.....	54
5.4 Data Storage	57
5.4.1 Admin Table	57
5.4.2 User Table.....	57
5.4.3 Health Data Table	58
5.4.4 Prediction Table.....	58
5.4.5 Medical History Table	59
5.4.6 Smart Health Tips Table	59
5.5 High Level Design	59
5.5.1 ER Diagram	59
5.5.2 Class Diagrams	61
5.6 Conclusion.....	62
CHAPTER 6: CONCLUSION AND FUTURE WORK	63
6.1 Future Work	63
6.2 Summary	63
REFERENCES.....	64

TABLE OF FIGURES

Figure 2-1: Pneumonia Disease	16
Figure 2-2: Viral Pneumonia	16
Figure 2-3: Fungal Pneumonia.....	17
Figure 2-4: classes Normal and Pneumonia chest X-ray	19
Figure 2-5: CheXNet Detection	22
Figure 2-6: DeepBreath System.....	24
Figure 2-7: Qure.ai System	25
Figure 2-8: Lunit INSIGHT CXR System	26
Figure 2-9: Zebra Medical Vision System.....	27
Figure 3-1: Waterfall Model Phases	31
Figure 3-2: Project Plan	35
Figure 3-3: Gantt Chart	36
Figure 4-1: Admin Use Case Diagram.....	44
Figure 4-2: User Use Case Diagram	45
Figure 4-3: Technical Use Case Diagram	46
Figure 4-4: Admin Sequence Diagram	47
Figure 4-5: User Sequence Diagram	48
Figure 4-6: Technical Sequence Diagram.....	49
Figure 4-7: Admin Activity Diagram	50
Figure 4-8: User Activity Diagram	51
Figure 4-9: Technical Activity Diagram.....	52
Figure 5-1: Welcome Interface	54
Figure 5-2: Register Interface	55
Figure 5-3: Login Interface	56
Figure 5-4: Dashboard Interface	56
Figure 5-5: Medical History Interface	56
Figure 5-6: ER Diagram.....	60
Figure 5-7: Class Diagram	61

LIST OF TABLES

Table 2-1: Comparison with similar systems.....	28
Table 3-1: Risk Management Plan for our project.....	32
Table 3-2: Tasks of our project	34
Table 4-1: Economic Feasibility	38
Table 4-2: Software Requirements	40
Table 4-3: Hardware Requirements	41
Table 5-1: Admin Table.....	57
Table 5-2: User Table	57
Table 5-3: Health Data Table.....	58
Table 5-4: Prediction Table	58
Table 5-5: Medical History Table.....	59
Table 5-6: Smart Health Tips Table.....	59

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
UML	Unified Modeling Language
CNN	Convolutional neural network
SVM	Support Vector Machines
K-NN	k-Nearest Neighbors

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Pneumonia is a life-threatening respiratory infection that affects millions of people worldwide each year, particularly young children, the elderly, and individuals with weakened immune systems. Early detection of pneumonia is crucial for timely treatment and the prevention of severe complications or mortality. Traditional diagnostic methods, such as physical examinations, laboratory tests, and chest X-rays, are often time-consuming and prone to human error, especially in settings with limited medical resources [1].

With the advancement of Artificial Intelligence (AI) and deep learning, the healthcare industry has witnessed significant transformations in diagnostic practices [2]. AI systems have shown remarkable potential in analyzing medical images, detecting abnormalities, and assisting healthcare professionals in making faster and more accurate decisions. Leveraging these technologies can reduce diagnostic delays, enhance medical efficiency, and improve patient outcomes [3].

Thus, we proposed MedPredict AI is developed as diagnostic app that integrates CNN deep learning model with an intuitive user interface. By analyzing chest X-ray scans alongside essential patient data, The system is designed to be accessible and beneficial to audience in achieving early pneumonia detection.

1.2 Problem Statement

Despite advances in medicine, pneumonia remains one of the leading causes of hospitalizations and deaths worldwide. In many regions, especially developing countries, access to qualified radiologists and diagnostic resources is limited, leading to delays in accurate diagnosis. Misinterpretation of chest X-rays or reliance solely on clinical symptoms can result in misdiagnosis, late treatment, or unnecessary medication [1].

Furthermore, existing diagnostic systems are often complex and not user-friendly, making them less accessible to patients and caregivers. There is a pressing need for a reliable, accessible, and intelligent platform that can support healthcare professionals by improving diagnostic accuracy while being easy to use.

1.3 Objectives of the project

General Objective:

To develop an AI diagnostic platform that utilizes deep learning techniques for accurate and efficient detection of pneumonia from chest X-ray images.

Specific Objectives:

- 1) To develop a deep learning model capable of accurately detecting pneumonia from medical image data.
- 2) To integrate both image-based and clinical input data for disease classification, enhancing diagnostic reliability.
- 3) To evaluate and validate the performance of the developed model using standard dataset.
- 4) To design a user-friendly app that allows medical professionals, patients, and caregivers to easily input data, upload scans, and access diagnostic results.
- 5) To contribute to the healthcare sector by providing a scalable, accessible, and efficient diagnostic solution.

1.4 Research Questions

This project is guided by the following research questions:

- i. How can deep learning techniques improve the accuracy of pneumonia detection compared to traditional diagnostic methods?

- ii. To what extent can combining chest X-ray images with clinical patient data enhance the reliability of pneumonia diagnosis?
- iii. What user interface features can make the platform accessible and intuitive for both healthcare professionals and patients?
- iv. How can this system contribute to early intervention and improved patient outcomes?

1.5 Scope of the Study

The scope of our study is centered on the development of MedPredict AI - Predictive Intelligence Model for Detecting Pneumonia, an android application that integrates deep learning and image processing technologies to support early and accurate pneumonia diagnosis. The system focuses on using technologies such as deep learning and image processing within a user-friendly system.

The implementing system in our study is Predictive Artificial Intelligence Technique for Detecting Pneumonia inside Saudi Arabia, and our application is free due to developed by students at King Khalid University as a graduation project.

1.6 Significance of the Study

The significance of our project holds significance in several dimensions:

- i. For Healthcare Professionals: Provides an intelligent assistant to support accurate diagnoses, especially in resource-limited areas.
- ii. For Patients and Caregivers: Offers an accessible tool for early disease detection, promoting timely treatment and better health outcomes.
- iii. For Medical Research: Demonstrates the effectiveness of deep learning in disease diagnosis, contributing to ongoing advancements in AI healthcare solutions.

- iv. For Society: Reduces the burden of pneumonia-related morbidity and mortality by enabling early detection.

1.7 Project Contribution

This research has the following contributions:

- i. Contribution 1: Patient History and Timeline Tracking

The system introduces a unique feature that compares multiple chest X-rays taken at different times for the same patient. This enables tracking of disease progression (improvement or deterioration), which most existing studies do not provide.

- ii. Contribution 2: Severity Based Risk Rating

Unlike conventional models that provide a binary result (pneumonia or not), our system generates a severity score and translates it into practical categories (mild, moderate, severe). This helps users better understand the urgency of the condition.

- iii. Contribution 3: Smart Health Tips

An App not only detects pneumonia but also offers AI generated health guidance. If the result suggests infection, the system provides initial advice such as seeking medical attention, maintaining hydration, or monitoring symptoms, bridging the gap between diagnosis and early care.

1.8 Organization of the Report

This report is structured as follows:

- o **Chapter 1: Introduction** – presents the introduction, including background, problem statement, objectives, research questions, scope, significance, contributions, and the report's organization.

- **Chapter 2: Literature Review** – reviews related literature and existing research on AI in medical imaging and pneumonia detection and comparative study.
- **Chapter 3: Project Management** – describes the project Plan, Gantt-chart, and Risk Management.
- **Chapter 4: Requirements and Analysis** – describes the methodology, system requirements, and system models used in the project.
- **Chapter 5: System Design** – presents the design process of the MedPredict AI platform including design choices for the system, including high level design, data storage, and user.
- **Chapter 6: Conclusion and Future Work** – concludes the project with findings, contributions, and future work for next semester(II).

1.9 Conclusion

This chapter introduced the project's background, motivation, and objectives, highlighting the importance of early pneumonia detection using AI. It established the problem context and outlined the system's expected contribution to healthcare.

CHAPTER 2: LITERATURE REVIEW

2.1 BACKGROUND

Pneumonia is a respiratory disease characterized by inflammation of the lung tissue, typically resulting in the alveoli filling with fluid or pus. This condition can lead to symptoms such as cough, fever, chest pain, and difficulty in breathing. Despite advances in healthcare, pneumonia remains a major cause of death globally, especially among vulnerable populations such as children under five years of age, elderly individuals, and immunocompromised patients. According to the World Health Organization (WHO, 2023), pneumonia accounts for approximately 15% of all deaths in children under five worldwide, highlighting the urgent need for improved diagnostic and preventive solutions [4].

Early diagnosis plays a critical role in the management of pneumonia. Traditional diagnostic approaches include physical examinations, chest X-rays, and laboratory tests. However, these methods depend on the availability of medical experts and diagnostic infrastructure, which may be scarce in rural or underdeveloped regions. Inaccurate or delayed diagnosis can result in disease progression, prolonged hospital stays, or even fatalities. This has driven research toward technology-based approaches such as Artificial Intelligence (AI) to support healthcare providers in improving diagnostic accuracy and efficiency.

2.1.1 Types of Pneumonia

Pneumonia is not a single disease but rather a condition with multiple causes and forms. Understanding its types is crucial for appropriate treatment:

- **Bacterial Pneumonia:**

Bacterial pneumonia is one of the most common and severe types of pneumonia, primarily caused by *streptococcus pneumoniae*, although other bacteria such as *haemophilus influenzae* or *staphylococcus aureus* can also be responsible. It typically develops suddenly and progresses rapidly, often presenting with high fever, chills, productive cough with thick phlegm, chest pain, and difficulty breathing. This form of pneumonia is usually more severe than other types and requires prompt

treatment with antibiotics to prevent complications such as sepsis or respiratory failure. As shown in Figure 2-1 from a diagnostic perspective, bacterial pneumonia is often identifiable on chest X-rays by dense, localized areas of lung consolidation, making it a suitable target for artificial intelligence (AI) based detection and classification systems in medical imaging [4].

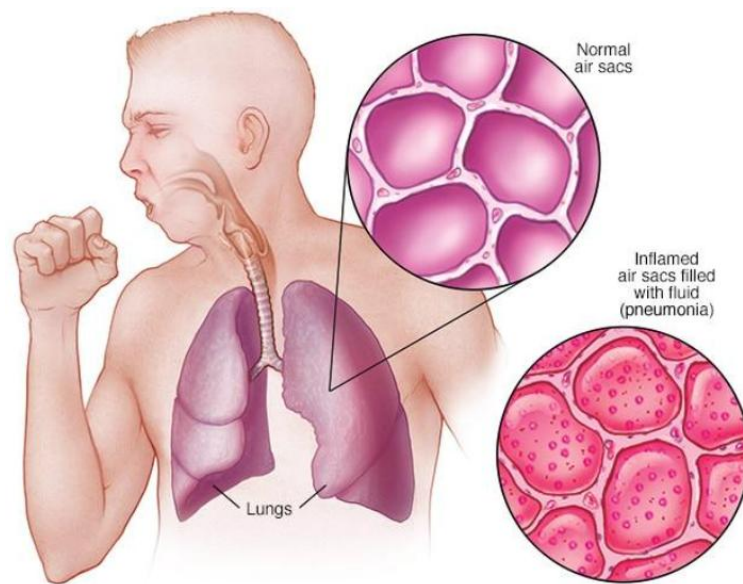


Figure 2-1: Pneumonia Disease

- **Viral Pneumonia:**



Figure 2-2: Viral Pneumonia

As shown in Figure 2-2, Viral pneumonia is caused by a variety of respiratory viruses, including influenza viruses, respiratory syncytial virus (RSV), adenoviruses, and more recently, coronaviruses such as SARS-CoV-2 (COVID-19). The symptoms often resemble those of bacterial pneumonia fever, cough, fatigue, and shortness of breath but tend to be milder and develop more gradually. Because viruses do not respond to antibiotics, treatment for viral pneumonia focuses on supportive care, such as rest, hydration, and antiviral medications in certain cases. Despite its usually less severe nature, viral pneumonia can still lead to significant complications, especially in young children, the elderly, and individuals with weakened immune systems. Chest imaging may show diffuse, patchy infiltrates rather than localized consolidation, which can help distinguish it from bacterial forms in clinical and AI based diagnostic systems [4].

- **Fungal Pneumonia:**

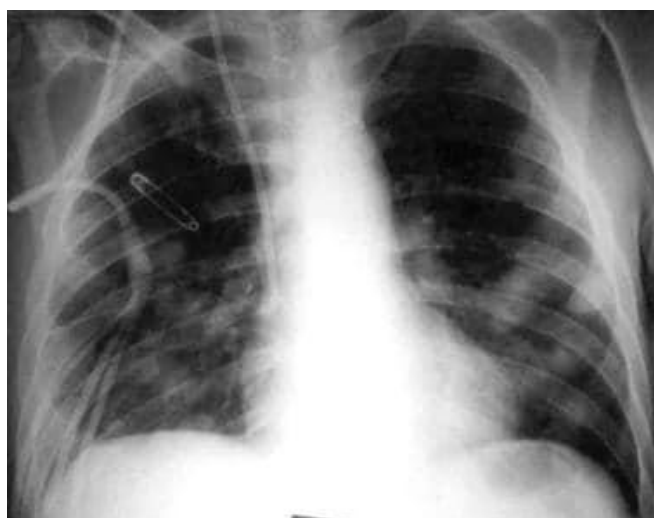


Figure 2-3: Fungal Pneumonia

As shown in Figure 2-3, Fungal pneumonia is one of the less frequent types of pneumonia that can be fatal, the fungi such as *Pneumocystis jirovecii*, *Histoplasma capsulatum*, *Aspergillus*, or *Cryptococcus* are the reason for this disease. It is an opportunistic infection that targets people with weakened immune systems such as patients with HIV/AIDS, cancer patients who are on chemotherapy, or organ transplant recipients who are taking immunosuppressive therapy. The symptoms are usually slow in their development and may consist of fever, dry cough, weight loss, and progressive shortness of breath. Fungal infections are difficult to diagnose because they may look like other types of pneumonia in the lungs by X-ray, but they

often show diffuse interstitial patterns or nodular infiltrates. The treatment that should be given is antifungal medications and it is very important that the disease is detected early to have a chance for successful therapy [4].

- **Community-Acquired Pneumonia (CAP):**

Community-acquired pneumonia is a term for infections that are picked up outside-of-hospital/health care settings and is the leading type of pneumonia on the planet. It can be a result of bacteria, viruses, or fungi, and *Streptococcus pneumoniae* is the most common bacterial cause. CAP can affect any person but is a big threat to the life of a baby, a geriatric person, and a patient with long-term illnesses. The symptoms can range from fever, cough, and chest pain to confusion and in case of severe difficulty in breathing. Diagnosis in most cases is based on the clinical evaluation, chest imaging, and laboratory tests, while the treatment depends on the pathogen of the disease. Since CAP is so common and diverse, the correct identification of its cause is the main stay for giving the right treatment and decreasing the rates of morbidity and mortality [4].

- **Hospital-Acquired Pneumonia (HAP):**

Hospital-acquired pneumonia is the infection that appears 48 hours or more after a patient has been admitted to a hospital or any other health care facility and was not there at the time of admission. It is typically linked to multidrug-resistant bacteria like *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, or Methicillin-resistant *Staphylococcus aureus* (MRSA), thus making the treatment more difficult than that of community-acquired infections. HAP is usually found among patients with compromised immune systems, those who have undergone invasive procedures, or prolonged hospitalized individuals. The signs might be severe and quick to advance, frequently resulting in serious side effects such as sepsis or respiratory failure. Because of its link to antibiotic-resistant organisms, the correct antibiotic therapy together with rapid diagnosis is of utmost importance [5].

- **Ventilator-Associated Pneumonia (VAP):**

Ventilator-associated pneumonia is a type of hospital-acquired pneumonia that is characterized by the development of the disease in patients who have been placed on mechanical ventilation through an endotracheal or tracheostomy tube, usually

after 48 hours of intubation. VAP is frequently caused by drug-resistant pathogens and is linked to high rates of morbidity, mortality, and longer hospital stays. The contributing factors for the development of the disease are extended ventilation, weakened host defenses, and the presence of aspiration of contaminated secretions. Symptoms of disease are fever, increased respiratory secretions, decline in oxygenation, as well as new or progressive lung infiltrates that can be seen on imaging. Due to its complexity and severity, VAP prevention requires the implementation of infection control measures, and management is usually with broad-spectrum antibiotics that are tailored according to culture results [5].

2.1.2 Role of Imaging in Pneumonia Diagnosis

Chest X-rays remain the gold standard for pneumonia diagnosis. They allow visualization of lung infiltrates, fluid buildup, and abnormal patterns indicative of infection. However, X-ray interpretation relies heavily on radiologist expertise, errors may occur due to similarities with other lung diseases (tuberculosis, lung cancer), in resource-limited areas, there may be a shortage of radiologists, delaying diagnosis. Thus, this gap in healthcare delivery provides an opportunity for AI systems to enhance diagnostic support [5].

2.1.3 Artificial Intelligence in Healthcare

Artificial Intelligence, particularly Deep Learning has revolutionized medical image analysis. Convolutional Neural Networks (CNNs) are the primary deep learning architectures used in pneumonia detection because of their ability to automatically extract hierarchical features from images. As shown in Figure 2-4 illustrates two chest X-ray image classes Normal and Pneumonia to distinguish healthy lungs from infected ones.

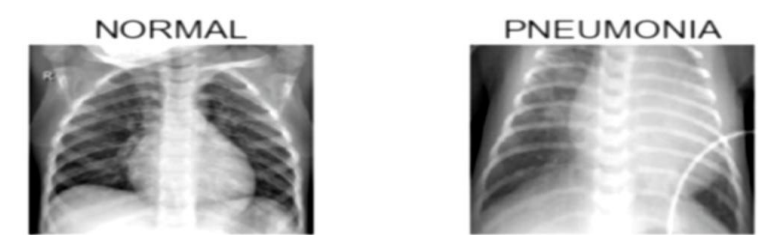


Figure 2-4: classes Normal and Pneumonia chest X-ray

Applications of AI in pneumonia detection include:

- Image Classification: Identifying whether a chest X-ray shows pneumonia or not.
- Severity Assessment: Assigning a risk score (mild, moderate, severe) for clinical decision-making.
- Integration with Clinical Data: Enhancing image-based diagnosis by incorporating patient history, symptoms, and demographic data.
- Advantages of AI-driven approaches:
 - Faster and more consistent results compared to manual interpretation.
 - Reduction of diagnostic errors.
 - Scalability in regions with limited medical resources.

2.1.4 Importance of Early Detection

Early detection of pneumonia significantly improves treatment outcomes. Benefits include:

- Reduced Mortality Rates: Especially in pediatric and elderly populations.
- Lower Healthcare Costs: Early treatment prevents hospitalizations and complications.
- Public Health Impact: Timely identification reduces the spread of infectious pneumonia.

2.2 History

The development of pneumonia detection using artificial intelligence (AI) has progressed through several important stages, beginning with traditional approaches and evolving into deep learning models.

2.2.1 Early Approaches: Traditional Image Processing and Machine Learning

In the early 2000s, research on automated pneumonia detection primarily relied on handcrafted image features such as edges, textures, and intensity values. These features were extracted from chest X-rays and processed using machine learning algorithms like Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Decision Trees. These early approaches provided the foundation for computational diagnosis and demonstrated the potential of automation, and the limitations in depended heavily on manual feature selection, lacked robustness, and often failed to generalize across diverse patient populations [1][3].

2.2.2 The Deep Learning Revolution

A major breakthrough occurred in 2012 when AlexNet, a deep Convolutional Neural Network (CNN), won the ImageNet competition, outperforming all traditional methods in image classification. This inspired researchers in medical imaging to adopt CNNs for disease detection, including pneumonia. Unlike traditional approaches, CNNs automatically learned features directly from raw images. This eliminated the need for manual feature engineering and improved diagnostic performance [2].

2.2.3 Growth of Medical Imaging Datasets (Mid-2010s)

The effectiveness of deep learning depends on access to large and diverse datasets. By the mid-2010s, publicly available medical image repositories, such as the NIH ChestX-ray14 dataset (comprising over 100,000 chest X-rays) and pediatric datasets from institutions like Guangzhou Women and Children's Medical Center, enabled researchers to train deep CNNs on a large scale, these datasets fueled a new wave of AI pneumonia detection research. They also highlighted challenges such as dataset imbalance, annotation errors, and variability across demographics [6].

2.2.4 Current Trends and Challenges (Present Day)

Today, pneumonia detection research focuses not only on accuracy but also on usability, explainability, and integration with healthcare systems. Attention is shifting toward pediatric pneumonia detection, as children present unique challenges. Remaining issues include dataset imbalance, interpretability of models, and ensuring fairness across populations [8].

2.3 SIMILAR SYSTEMS

2.3.1 CheXNet

CheXNet is one of the earliest and most widely cited deep learning models for pneumonia detection. Developed by researchers at Stanford University, it is a 121-layer convolutional neural network (CNN) based on DenseNet architecture. The system was trained on the NIH ChestX-ray14 dataset, which contains over 100,000 chest X-ray images annotated with 14 different thoracic diseases. As shown in Figure 2-5, CheXNet was particularly notable for achieving radiologist-level performance in pneumonia detection, marking a breakthrough in applying deep learning to medical imaging [6].

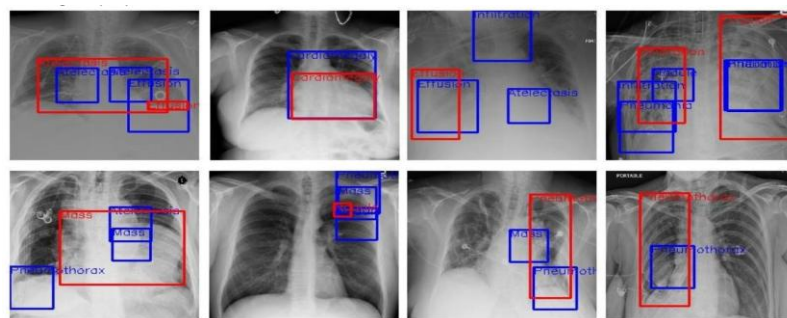


Figure 2-5: CheXNet Detection

Advantages:

- Achieved high accuracy in pneumonia detection.
- Leveraged a large public dataset, ensuring reproducibility.

- Provided the foundation for many subsequent AI healthcare studies.

Disadvantages:

- Requires high computational resources for training and deployment.
- Lacks interpretability, making clinical trust challenging.
- Not Free.

2.3.2 COVID-Net

In this research COVID-Net is a deep learning model introduced during the COVID-19 pandemic to assist in detecting COVID-19 infections from chest radiographs. Developed by researchers in Canada, it was designed as an open-source neural network tailored specifically for COVID-19 detection. The model was trained using a combination of datasets, including normal chest X-rays, pneumonia cases, and COVID-19 cases, allowing it to distinguish between them effectively [7].

Advantages:

- Specifically designed for emerging respiratory diseases, including pneumonia.
- Provided open-source models and datasets for global collaboration.
- Included explainability methods to improve trust in results.

Disadvantages:

- Initially limited by small and imbalanced datasets.
- Primarily focused on COVID-19, with less emphasis on general pneumonia.
- Performance varied across demographics and imaging devices.

2.3.3 DeepBreath System

DeepBreath is an AI system designed for respiratory disease detection, including pneumonia, using CNNs trained on multiple chest X-ray datasets. It emphasizes fast processing for near real-time screening in hospital settings, and attractive for use in emergency and low-resource healthcare environments where radiologists may not be immediately available [8]. As described in Figure 2-6, the DeepBreath system architecture demonstrates the workflow of inputting a chest X-ray, applying CNN-based feature extraction, and producing disease predictions in real-time.

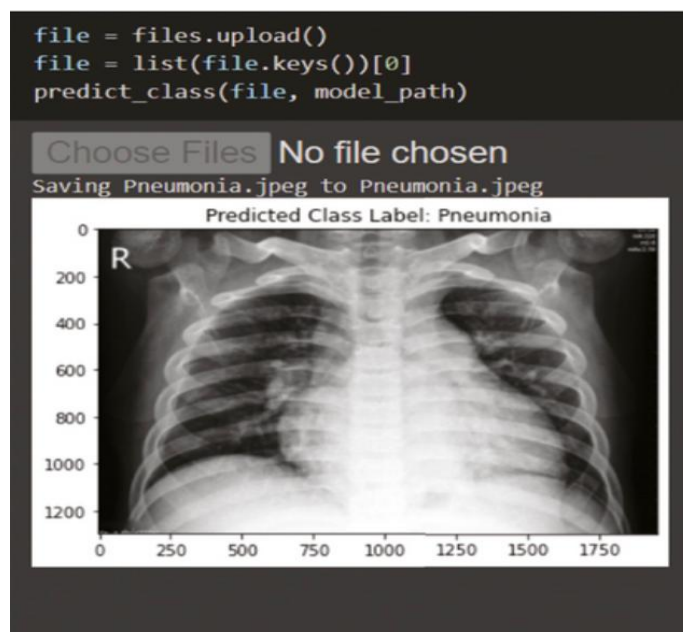


Figure 2-6: DeepBreath System

Advantages:

- Provided rapid diagnosis suitable for emergency use.
- Integrated multiple datasets to improve generalization.
- Designed for deployment in low-resource clinical environments.

Disadvantages:

- Limited peer-reviewed validation compared to CheXNet.
- It underperforms on rare pneumonia subtypes.

- Lacks integration with patient clinical history.

2.3.4 Qure.ai qXR System

Qure.ai's qXR system is a commercially available AI-powered chest X-ray interpretation tool. Unlike research-only systems, qXR has been deployed in hospitals and clinics worldwide, particularly in low-resource settings where radiologists are scarce. The system provides automated reports within seconds, making it highly valuable in time-sensitive environments. Additionally, while qXR offers high accuracy, it does not provide patient-focused features such as history tracking or preventive health tips [9]. As shown in Figure 2-7 displays the Qure.ai qXR system interface, showing automated detection of pneumonia regions and result summaries integrated into the hospital's workflow.

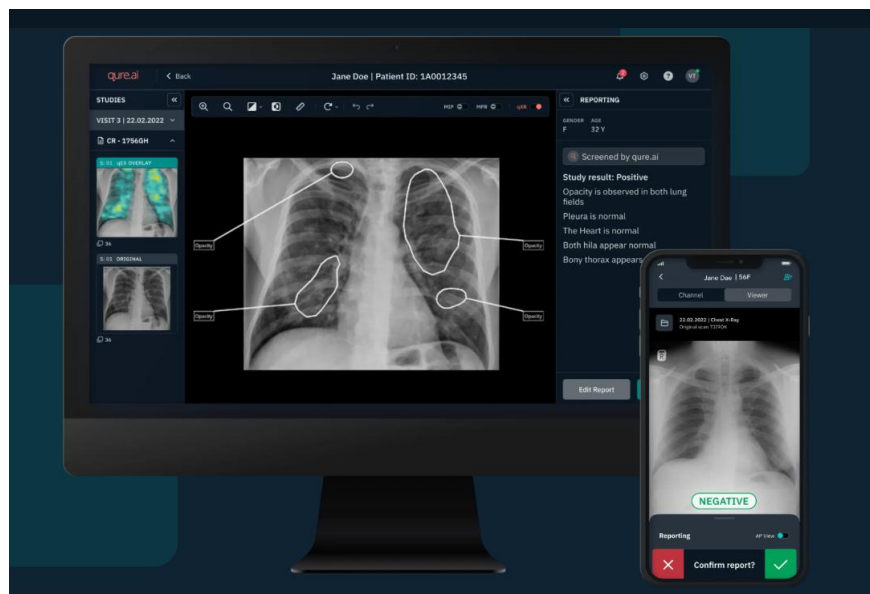


Figure 2-7: Qure.ai System

Advantages:

- Clinically validated and used in real-world healthcare settings.
- Detects multiple abnormalities beyond pneumonia.
- User-friendly interface suitable for clinical workflows.

Disadvantages:

- Proprietary system with limited transparency.
- Not Free.
- Face generalization issues across low-quality datasets.

2.3.5 Lunit INSIGHT CXR System

Lunit INSIGHT CXR is an AI chest X-ray analysis software in South Korean medical AI company. The system is designed to assist radiologists by detecting and highlighting abnormal findings such as pneumonia, tuberculosis, nodules, and other thoracic diseases. It integrates seamlessly with existing hospital PACS (Picture Archiving and Communication System), allowing radiologists to receive instant AI feedback during image interpretation [10]. As shown in Figure 2-8, the Lunit INSIGHT interface highlights abnormal lung areas detected by the AI, supporting rapid and clinical decisions.

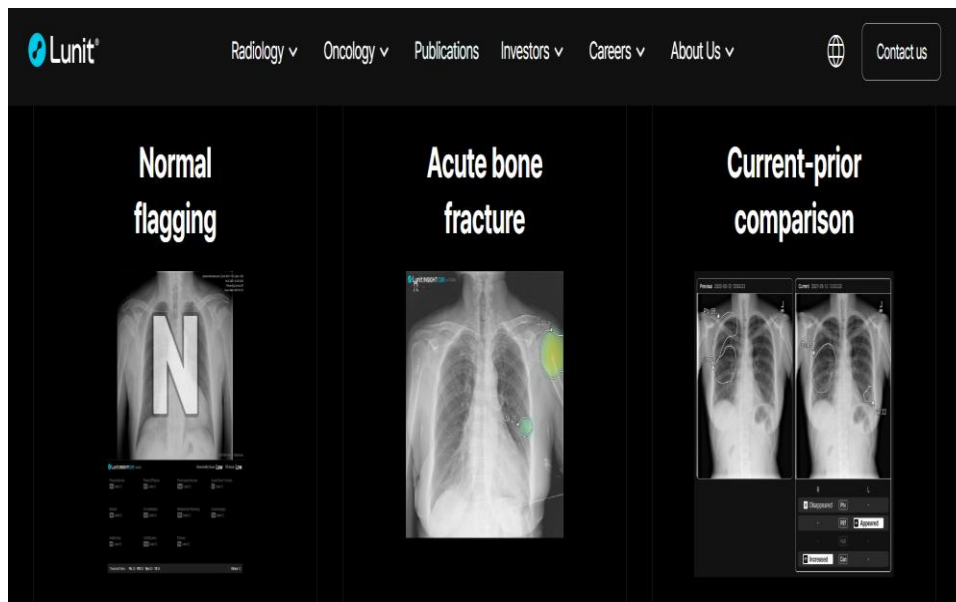


Figure 2-8: Lunit INSIGHT CXR System

Advantages:

- FDA-approved and CE-certified, ensuring reliability and clinical acceptance.
- Provides visualization of detected abnormalities, increasing transparency.
- Integrates easily with hospital PACS systems.

Disadvantages:

- Commercial and expensive, limiting accessibility in low-resource settings.
- Requires strong internet and system infrastructure for deployment.
- Primarily designed for hospital use, not easily accessible for individual patients.

2.3.6 Zebra Medical Vision System

Zebra Medical Vision medical imaging AI company developed an AI system that analyzes chest X-rays to detect conditions such as pneumonia, emphysema, and lung nodules. Their system is cloud-based and designed to provide scalable AI services for hospitals and radiology centers. Zebra-Med's system not only detects pneumonia but also integrates with multiple imaging modalities to analyze CT scans, mammograms, and bone density scans [11]. As shown in Figure 2-9, the system's dashboard presents detection outputs for multiple thoracic diseases across diverse imaging data.

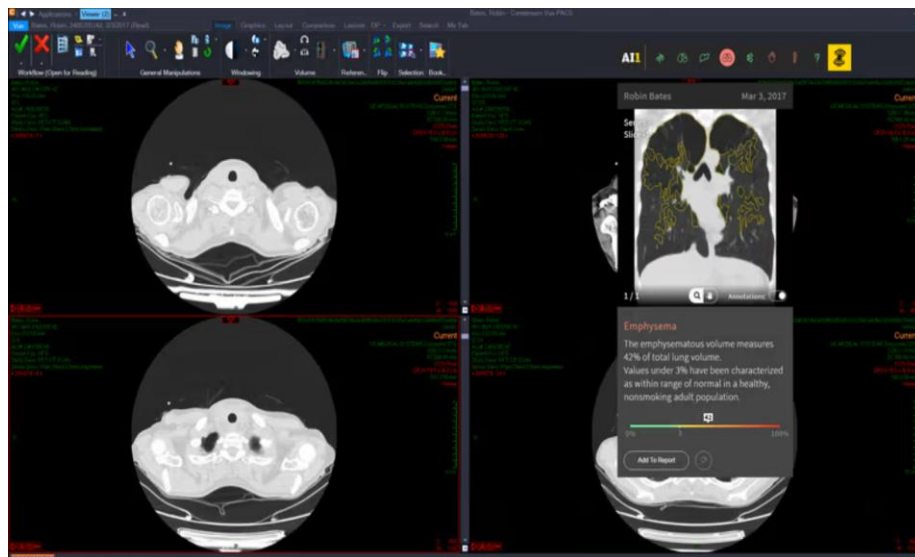


Figure 2-9: Zebra Medical Vision System

Advantages:

- Multi-disease detection capability across different imaging modalities.
- Cloud-based, making it scalable and accessible to multiple healthcare providers.
- Offers population-level screening for public health and early detection programs.

Disadvantages:

- Heavy reliance on cloud infrastructure may raise concerns about data privacy.
- Not Free.
- Requires stable internet connectivity, making it less practical in low-resource settings.

2.3 Comparative Study

In this section, we present a comparative study of existing pneumonia detection systems. Each system is analyzed based on its functionality, and unique features. As shown in Table 2-1, the comparison summarizes similarities and differences among well-known systems such as DeepBreath, Qure.ai qXR, Lunit INSIGHT, and Zebra Medical Vision, highlighting how our proposed system improves on current limitations.

Table 2-1: Comparison with similar systems

Features	Our System	CheXNet	COVID-Net	DeepBreath System	Qure.ai qXR	Lunit INSIGHT CXR	Zebra Medical Vision
Patient History & Timeline Tracking	Yes	No	No	No	No	No	Partial
Severity & Risk Rating	Yes (mild, moderate, severe)	No (binary output)	No (infection-focused)	No (breathing metrics)	Partial (confidence levels)	No	Partial (confidence levels)
Multi-Modal Input (Images + Clinical Data)	Yes	No	No	No	Partial	Partial	Yes
Smart Health Tips	Yes	No	No	No	No	No	Yes
Cost	Free	Not Free	Free	Not Free	Not Free	Not Free	Not Free
AI Model	Custom CNN	DenseNet-121	COVID-Net	Custom CNN / Transfer Learning	Proprietary Deep Learning Model	Proprietary Deep Learning Model	Proprietary Deep Learning Model

2.4 Conclusion

This chapter summarized the findings from existing literature, current systems, and research trends in medical image classification. It identified limitations in previous work, and comparative study between our project and similar systems.

CHAPTER 3: PROJECT MANAGEMENT

3.1 Introduction

In this chapter we outline how the project was systematically planned, organized, and executed to achieve the intended goals efficiently and within the given timeframe. Effective project management helps the team allocate resources properly, manage risks, and track progress. The project plan was developed by considering both technical and academic requirements, aligning with the team's schedule and deliverables set by department of informatics & computer systems.

3.2 Approach

The approach selected for the development of our project is based on the Waterfall Software Development Life Cycle (SDLC) Model. This model was chosen because it provides a structured and sequential process that ensures clarity and organization in each phase of the project. Since our system requires careful planning, medical data handling, and validation, the Waterfall model allows us to move systematically through well-defined stages minimizing risks and ensuring high quality in the final system [12].

3.2.1 Justification for Selecting the Waterfall Model:

- Clear and Sequential Structure: Each stage in the Waterfall model (Requirements, Design, Implementation, Testing, and Maintenance) follows a logical order, making it ideal for academic and research-based projects that require documentation and systematic progress tracking.
- Well-defined Requirements: The system's requirements such as image processing, patient history tracking, and severity scoring are clearly identified at the beginning of the project, allowing for precise planning and execution.
- Ease of Project Management: The Waterfall model facilitates monitoring and control, since every stage has specific deliverables and review points. This helps the team to meet deadlines efficiently and manage resources effectively.

3.2.2 Phases of Waterfall Model

The development of our project follows the Waterfall model, a structured approach that divides the process into sequential phases where each phase depends on the output of the previous one. In this project, Figure 3-1 illustrates the major phases of the Waterfall Model, including Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance [12].

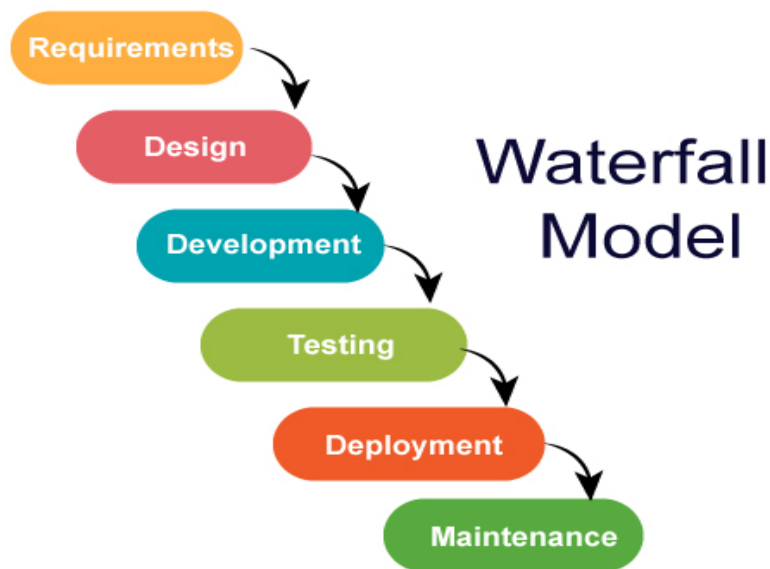


Figure 3-1: Waterfall Model Phases

- 1) Requirement Analysis: Gathering and analyzing user and system requirements (input data, X-ray images, output reports, etc.).
- 2) System Design: Designing system architecture, database structure, and user interface.
- 3) Implementation: Developing an android application using android studio using java language, deep learning model, image processing tools, technologies(TensorFlow lite), and firebase DB for data storage.
- 4) Testing: Verifying model accuracy, validating performance, and ensuring the system functions correctly.
- 5) Deployment: Launching an app platform and ensuring it operates smoothly for end users as APK android application.
- 6) Maintenance: Addressing any bugs, updating the AI model, and improving features based on feedback.

3.3 Risk Management

Risk management is an essential part of our project to anticipate, identify, and mitigate potential challenges during the development process. The project team identified possible risks in technical, operational, and scheduling areas, along with plans to minimize their effects. As presented in Table 3-1, the risk management plan outlines possible risks, their likelihood, impact, and corresponding mitigation measures.

Table 3-1: Risk Management Plan for our project

Risk Description	Type of Risk	Likelihood	Impact	Mitigation Strategy
Functional Risk Management				
Dataset Quality Issues – The medical X-ray dataset may contain low-resolution or mislabeled images, leading to inaccurate training results.	Technical	High	High	Use publicly verified datasets (Kaggle Pneumonia Dataset); perform preprocessing and validation before training the model.
AI Model Overfitting – The CNN model may perform well on training data but poorly on unseen test data.	Technical	Medium	High	Implement dropout layers, data augmentation, and k-fold cross-validation; monitor model generalization accuracy.
Data Privacy and Security Risks – Handling medical data can expose privacy vulnerabilities.	Security	Medium	High	Apply encryption, anonymize patient data.
Non-Functional Risk Management				
Poor Time Management.	Management	High	High	Follow the Gantt chart strictly, hold weekly progress meetings, and assign clear task ownership.

Tools or Software Compatibility – Issues may arise between development tools, such as Python libraries and web frameworks.	Technical	Medium	Medium	Test software environments early and standardize tool versions.
Team Communication – Miscommunication or lack of coordination among members can cause duplication or delays.	Organizational	Medium	High	Use communication platforms (WhatsApp, Google Meet) for consistent updates and assign a team leader for coordination.
Hardware Limitation – AI model training and image processing may exceed available computational resources.	Technical	Medium	High	Use cloud-based resources such as Google Colab or Kaggle GPU environments.

3.4 Project Plan

The project plan for our project predictive artificial intelligence technique for detecting pneumonia outlines the major tasks and responsibilities distributed among team members. Table 3-2 provides the distribution of project tasks among team members, showing the structured approach to managing progress and collaboration.

Table 3-2: Tasks of our project

No.	Task	Team Members Responsible
1	Create team, determine project idea, programming language, objectives, and submit the project plan.	All Members
2	Write Chapter 1: Introduction (Background, Problem Statement, Objectives, Research Questions).	Lamis Mohammed, Renad Mohammed, Nada Abdulla
3	Write Chapter 1: Introduction (Scope of the Project, Significance of the Study, Project Contribution, Organization of the Report).	Arwa Salman, Taif Abdullah, Nada Abdulla
4	Collect necessary information and resources.	Lamis Mohammed, Taif Abdullah
5	Study the current systems, review related research, and compare the proposed system with existing ones.	All Members
6	Write and deliver Chapter 2: Background and Literature Review.	Renad Mohammed, Nada Abdulla, Arwa Salman
7	Determine in Chapter 3: Project Management, Approach, Risk Management, and Project Plan	Lamis Mohammed, Nada Abdulla
8	Determine and analyze system requirements (functional and non-functional), and Design system models (Use Case Diagram, Activity Diagram, Sequence Diagram).	Arwa Salman, Renad Mohammed, Taif Abdullah
9	Write and deliver Chapter 4: System Analysis.	All Members
10	Perform high-level design, create initial interface prototypes, and add descriptions for each.	Lamis Mohammed, Renad Mohammed, Nada Abdulla
11	Write and deliver Chapter 5: System Design.	Arwa Salman, Taif Abdullah
12	Prepare Chapter 6: Conclusion and Future Work.	Lamis Mohammed, Nada Abdulla

3.4.1 Plan and Gantt Chart

The overall timeline of the project is visualized using a project plan and Gantt chart to track progress and ensure timely delivery of each milestone. As shown in Figure 3-2, the project plan outlines the sequence of activities. Meanwhile, Figure 3-3 represents the Gantt Chart, providing a visual overview of task durations and dependencies across the entire project schedule.

	الاسم	المدة	البداية	النهاية
1	 Chapter 1: Introduction	5 days	ص 08:00 19/09/25	م 05:00 25/09/25
2	Background of the Study	1 day	ص 08:00 19/09/25	م 05:00 19/09/25
3	Problem Statement	1 day	ص 08:00 22/09/25	م 05:00 22/09/25
4	Objectives of the study	1 day	ص 08:00 23/09/25	م 05:00 23/09/25
5	Research Questions	0.5 days	ص 08:00 24/09/25	م 01:00 24/09/25
6	Scope of the Study	0.5 days	م 01:00 24/09/25	م 05:00 24/09/25
7	Significance of the study	0.5 days	ص 08:00 25/09/25	م 01:00 25/09/25
8	Project Contribution	0.5 days	م 01:00 25/09/25	م 05:00 25/09/25
9	Organization of the Report	0.5 days	م 01:00 25/09/25	م 05:00 25/09/25
10	 Chapter 2: Literature Review	9 days	ص 08:00 26/09/25	م 05:00 08/10/25
11	Introduction	1 day	ص 08:00 26/09/25	م 05:00 26/09/25
12	Similar Systems	4 days	ص 08:00 29/09/25	م 05:00 02/10/25
13	Comarative Study	4 days	ص 08:00 03/10/25	م 05:00 08/10/25
14	 Chapter 3: Project Management	3 days?	ص 08:00 09/10/25	م 05:00 13/10/25
15	Approach	1 day?	ص 08:00 09/10/25	م 05:00 09/10/25
16	Risk Management	1 day?	ص 08:00 10/10/25	م 05:00 10/10/25
17	Project Plan	1 day?	ص 08:00 13/10/25	م 05:00 13/10/25
18	 Chapter 4: Requirements and Analysis	5 days	ص 08:00 14/10/25	م 05:00 20/10/25
19	Introduction	1 day	ص 08:00 14/10/25	م 05:00 14/10/25
20	Methodology	1 day	ص 08:00 15/10/25	م 05:00 15/10/25
21	System Requirements	1 day	ص 08:00 16/10/25	م 05:00 16/10/25
22	Functional and non-functional	1 day	ص 08:00 17/10/25	م 05:00 17/10/25
23	System Models	2 days	ص 08:00 17/10/25	م 05:00 20/10/25
24	 Chapter 5: System Design	7 days	ص 08:00 21/10/25	م 05:00 29/10/25
25	Introduction	1 day	ص 08:00 21/10/25	م 05:00 21/10/25
26	High Level Design	3 days	ص 08:00 22/10/25	م 05:00 24/10/25
27	User Interface Design	3 days	ص 08:00 27/10/25	م 05:00 29/10/25
28	 Chapter 6: Conclusion and Future	3 days?	ص 08:00 10/11/25	م 05:00 12/11/25
29	Conclusion	1 day	ص 08:00 10/11/25	م 05:00 10/11/25
30	Future Work	1 day?	ص 08:00 11/11/25	م 05:00 11/11/25
31	References	1 day	ص 08:00 12/11/25	م 05:00 12/11/25

Figure 3-2: Project Plan

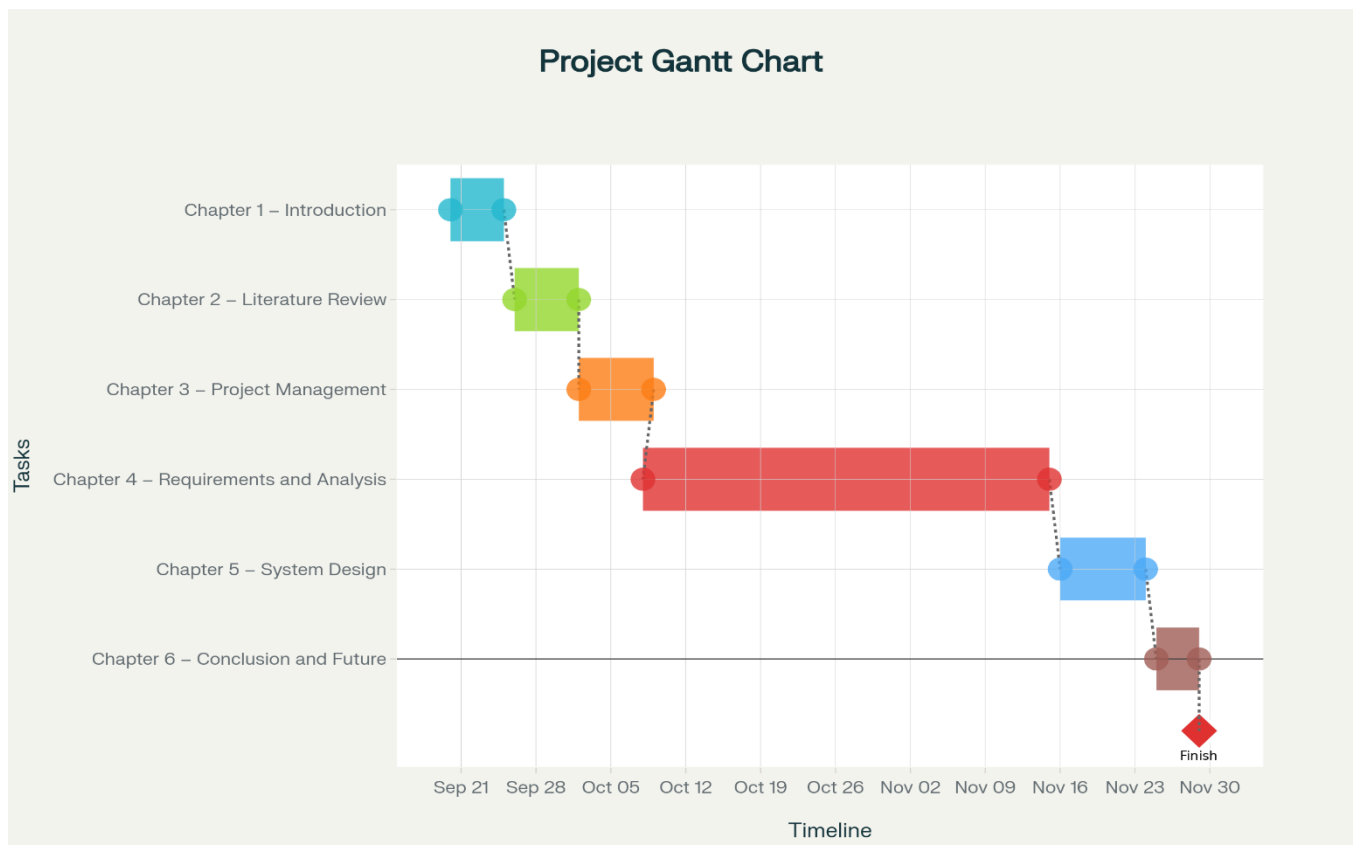


Figure 3-3: Gantt Chart

3.5 Conclusion

This chapter detailed the system's approach, methodology, and project management, including SDLC, risk management, and project plan and Gantt-chart. This chapter the basis for accurate implementation and system development in the next stages.

CHAPTER 4: REQUIERMENTS AND ANALYSIS

4.1 Introduction

In this section examines system analysis, which is the method of designing and defining a system's structure, parts, and operations. This design is developed according to specified functional and non-functional needs, necessary software and hardware, and relevant system models and diagrams.

4.2 Feasibility Study

A feasibility study is a crucial step in determining the viability and potential success of our project. It helps assess whether the project can be developed efficiently within the available resources, technology, and time frame [13].

4.2.1 Operational Feasibility

Operational feasibility determines whether the system will operate effectively within the healthcare environment. The proposed system will assist users by providing an automated pneumonia detection process from chest X-ray images. The system will reduce diagnostic time. Additionally, the interface will be designed to be user-friendly, ensuring ease of use without requiring deep technical expertise [13].

4.2.2 Technical Feasibility

Technical feasibility evaluates whether the existing technology is adequate for implementing the system. The project will be developed using Python, with frameworks such as TensorFlow and Keras for model training and Android Studio for deployment. The AI model will be trained on publicly available Chest X-Ray datasets. The required hardware and software are widely available, and the technologies are well-suited for deep learning-based medical image classification [13].

4.2.3 Economic Feasibility

The economic feasibility of the project assesses the financial requirements needed to develop and maintain the system. Since the project uses open-source tools and freely available datasets, the overall cost is minimal. Table 4-1 summarizes the estimated cost of system development [13].

Table 4-1: Economic Feasibility

Description	Unit Price
Hardware	4000
Software Cost (Android Studio)	Free trial
Communication Cost	500
Meeting Cost	Free
Total System Development Cost	4500 SAR

4.3 System Requirements

Requirements define what the system must accomplish and the constraints under which it must operate. They are divided into functional and non-functional categories to ensure that the system performs efficiently and reliably.

4.3.1 Functional Requirements

Functional requirements describe the specific tasks and services the system must perform. They define how the system should react to user inputs and behave under different conditions. These requirements are shaped by the software's purpose, its target users, and the operational environment. The functional requirements for our system are as follows:

4.3.1.1 Admin Functional Requirements:

- login/logout to the system.
- Manage user accounts (add, delete, update).
- Upload and manage dataset of deep learning model.
- View the prediction results of pneumonia.
- Manage system configuration and security settings.

4.3.1.2 User Functional Requirements:

- Register and log in to the system.
- Upload chest X-ray images for pneumonia detection.
- View prediction results and Risk Rating.
- Track medical history and timeline of previous prediction results.
- Access medical recommendations or reports generated by the system.
- Log out from the system.

4.3.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance standards of the system. The non-functional requirements that are relevant to our project:

- **Performance:** The system must classify each uploaded chest X-ray image within 5 seconds. To achieve this:
 - The deep learning model will be optimized using techniques such as model quantization, and Efficient preprocessing pipelines (TensorFlow I/O) will minimize delay.

- **Reliability:** The system must maintain 95% uptime, ensuring continuous availability for users and healthcare providers. technical measures include redundant backup storage for logs, predictions, and user data.
- **Usability:** The interface should be simple and intuitive for users. achieved through responsive UI design compatible with desktop and mobile devices, and clear upload buttons, results screens, and timelines.
- **Security:** All patient information and medical images uploaded must remain protected from unauthorized access. implementation includes:
 - End-to-end encryption (TLS/SSL) for data transmission.
 - AES-based encryption for stored medical images.

4.3.3 Software Requirements

The development and operation of the system rely on a carefully selected set of software tools and technologies. Each component in the stack serves a specific purpose, from building the web interface and processing data to managing information and creating design diagrams. The following 4-2 table details the tools used in our project:

Table 4-2: Software Requirements

Software Tool	Description
Android Studio	The primary integrated development environment (IDE) utilized for writing, testing, and debugging android application's source code.
Edraw-Max	A dedicated diagramming tool employed for designing UML diagrams and creating visual representations of the system's architecture and workflows.

ANACONDA (PYTHON)	A Python distribution that simplifies package management and environment isolation, crucial for handling deep learning libraries.
Firebase DB	Cloud Storage database management system that serves as the central repository for storing patient records, user data, and diagnostic results.

4.3.4 Hardware Requirements

To ensure optimal performance and stability, the system requires specific hardware configuration. The following 4-3 table outlines the minimum recommended hardware requirements:

Table 4-3: Hardware Requirements

Hardware	Description
CPU	Intel Processor Core-i7
RAM	8GB or more
HARD DISK	256 GB SSD

4.4 Personas

In this section describes the characteristics of the system's main users. Our system is designed to serve two primary user types, each with distinct roles and objectives that contribute to the functionality of the system.

1) User(Patient):

- Role: End-user who interacts with the system to detect pneumonia.
- Responsibilities: Uploads chest X-ray images for analysis and receives AI-generated detection results.
- Objectives: Obtain fast, accurate, and easy-to-understand diagnostic feedback and possible recommendations regarding pneumonia detection.
- Key Benefit: Enables non-expert users to access AI-driven diagnostic assistance remotely without the need for immediate medical consultation.

2. System Administrator:

- Role: Technical and operational maintainer of our application.
- Responsibilities: Manages user accounts, monitors system performance, updates the dataset, and maintains AI model accuracy through regular retraining and optimization.
- Objectives: Ensure system reliability, data security, and continuous improvement of AI model performance.
- Key Benefit: Maintains a stable, secure, and efficient system environment for accurate pneumonia detection.

3. Technical Service Provider (Developer):

- Role: External/back-end service provider enabling infrastructure and model deployment.
- Responsibilities:
 - Ensure availability, and data-handling performance of the AI model.
 - Provide technical support for AI model deployment, model updates, and integration tools.

- Handle backend security protocols (encryption, access control, model protection).
- Objectives: Guarantee stable infrastructure, fast predictions, and secure storage/processing of medical images.
- Benefit: Ensures the system operates smoothly behind the scenes, enabling scalable, high-performance AI services.

4.5 System Models

The Unified Modeling Language (UML) is used to visually represent system functionality and interactions between components. UML helps understand how users interact with the system and how different modules communicate.

4.5.1 Use Case Diagrams

Use case diagrams will visually represent the relationships between the system's actors—such as users and administrators—and the functions they perform. These diagrams will clarify the specific features and permissions allocated to each role within the system. As shown in figure 4-1 use case diagram of our project.

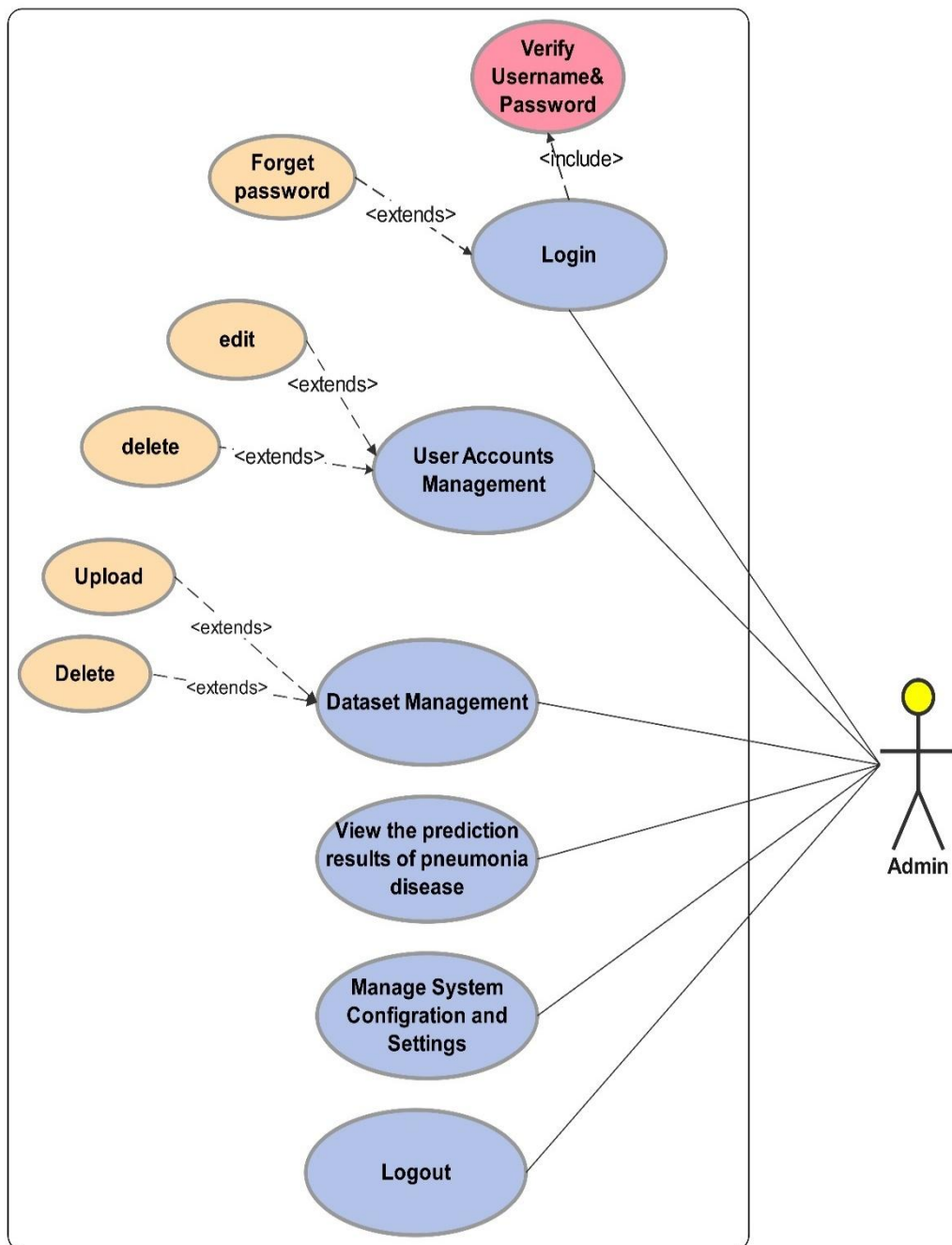


Figure 4-1: Admin Use Case Diagram



Figure 4-2: User Use Case Diagram

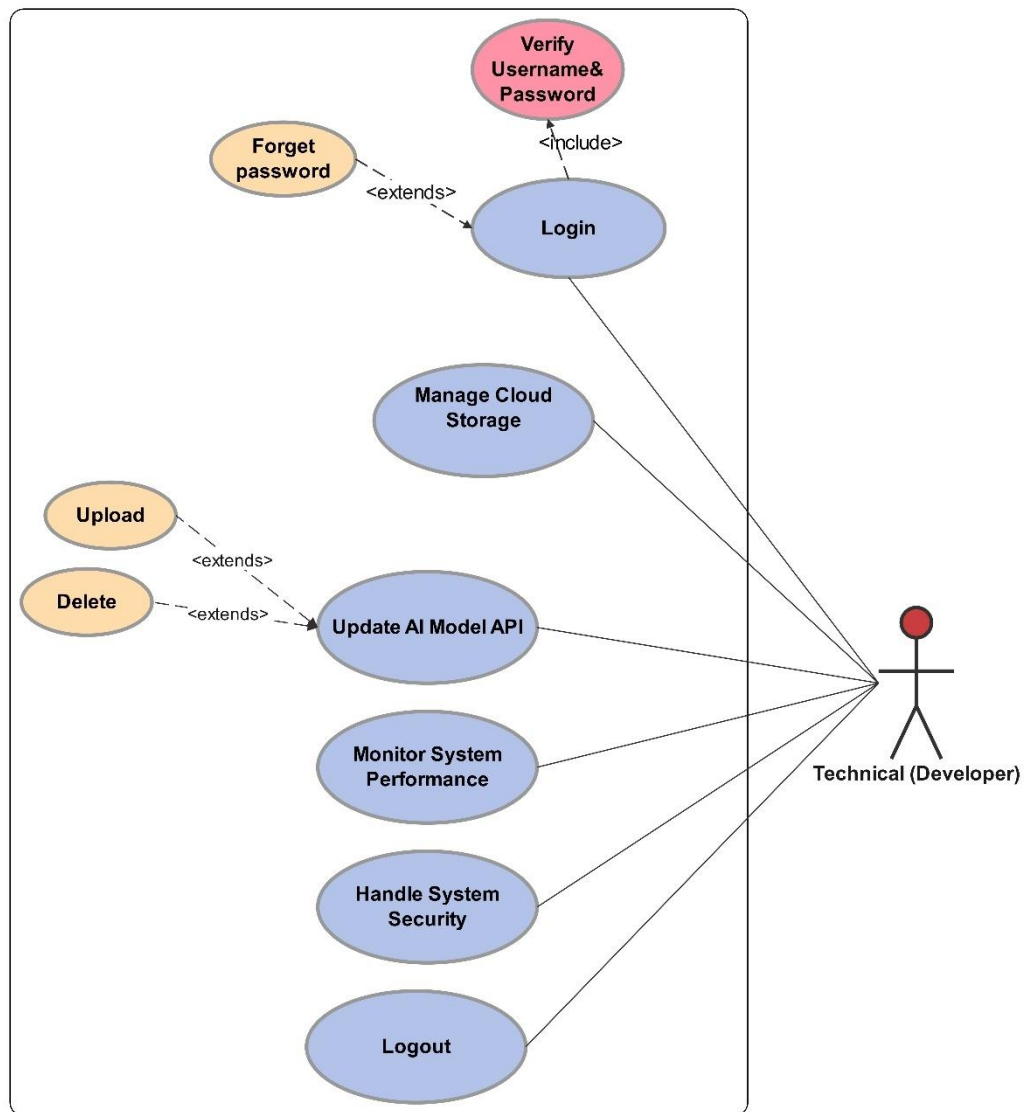


Figure 4-3: Technical Use Case Diagram

4.5.2 Sequence Diagram

Sequence diagrams will model the time-ordered sequence of interactions among system components, such as user interfaces, system logic, and server modules. They capture the detailed flow of messages and operations that occur during specific processes, as visualized in Figure 4-4.

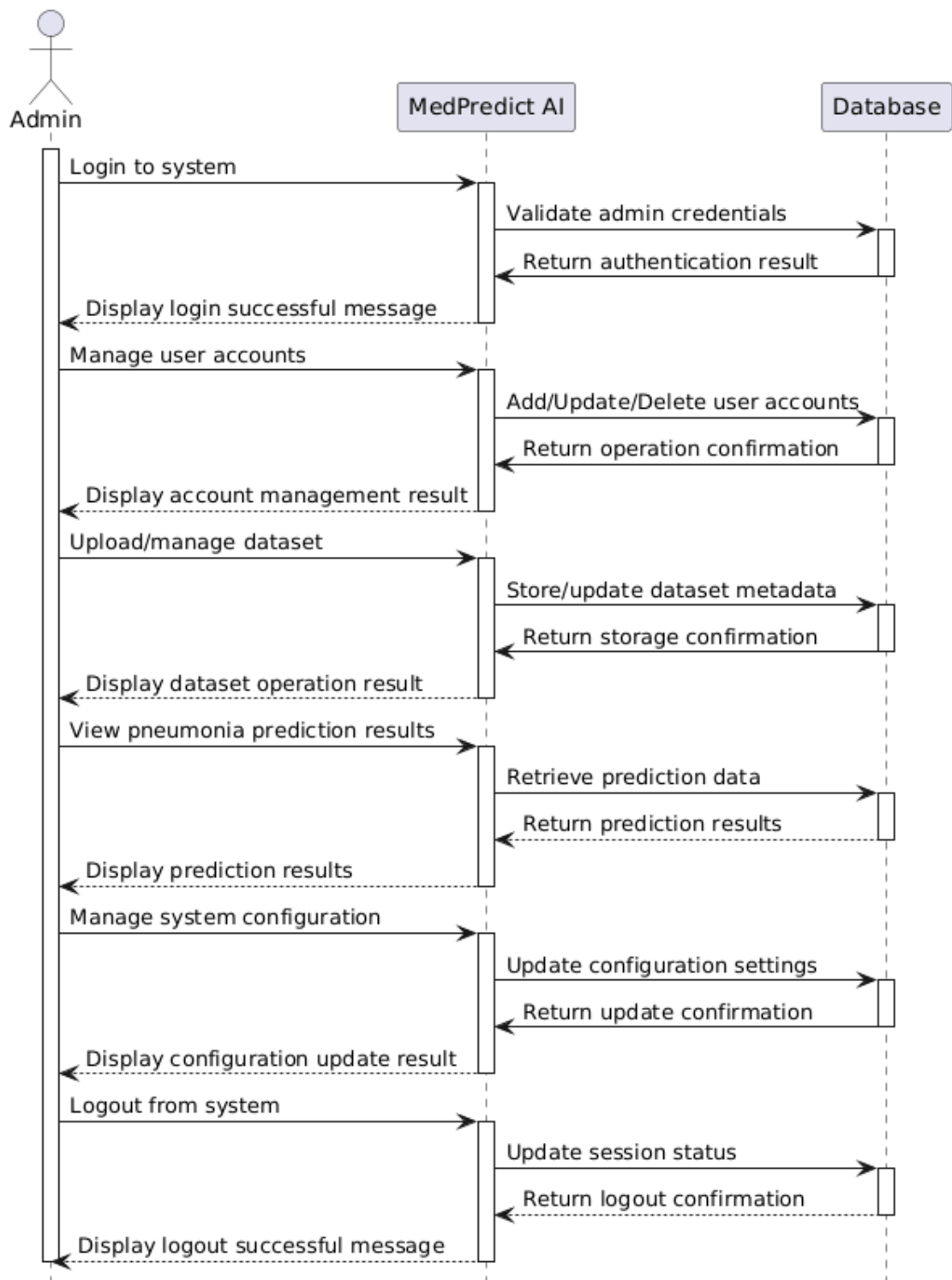


Figure 4-4: Admin Sequence Diagram

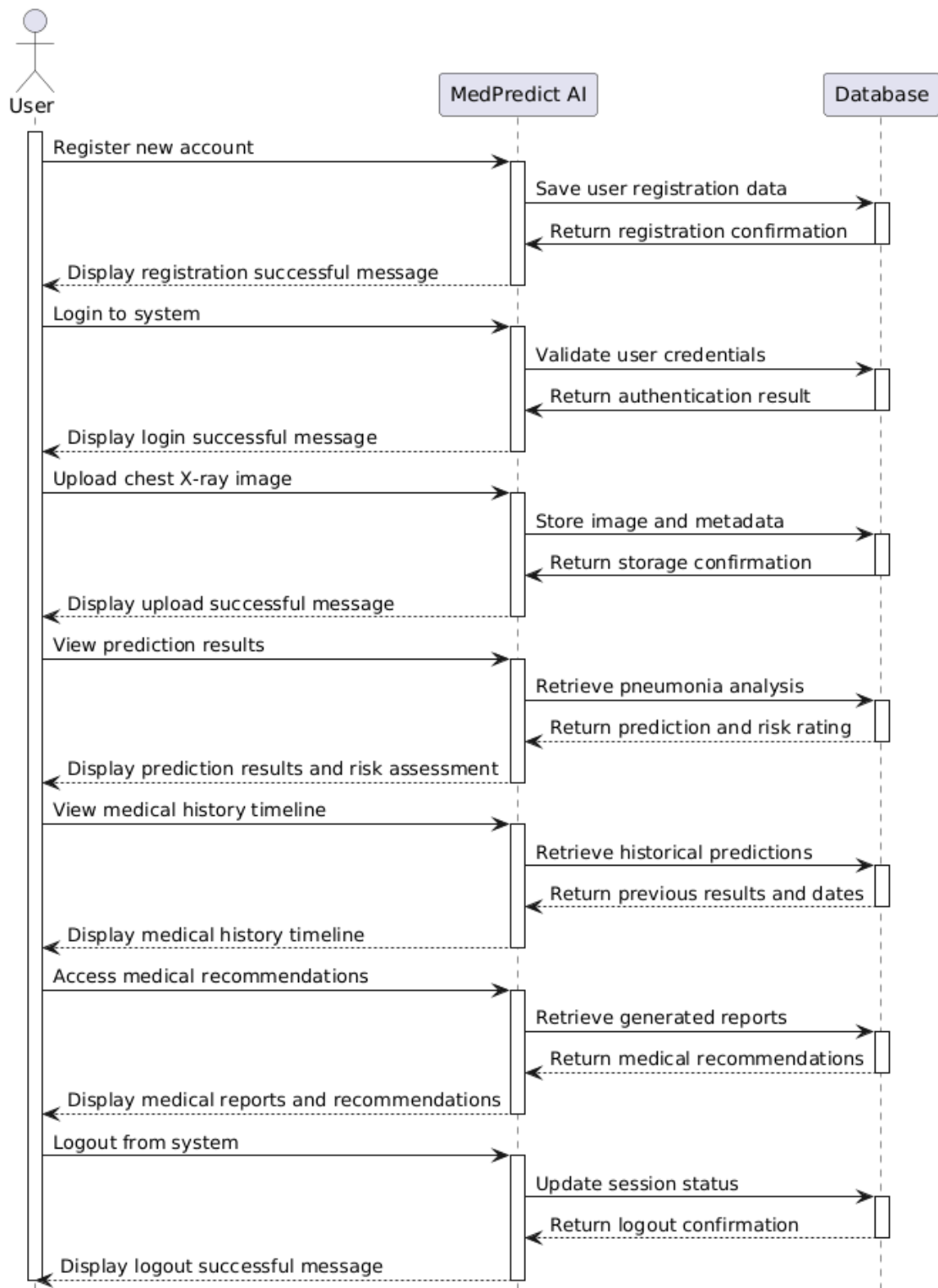


Figure 4-5: User Sequence Diagram

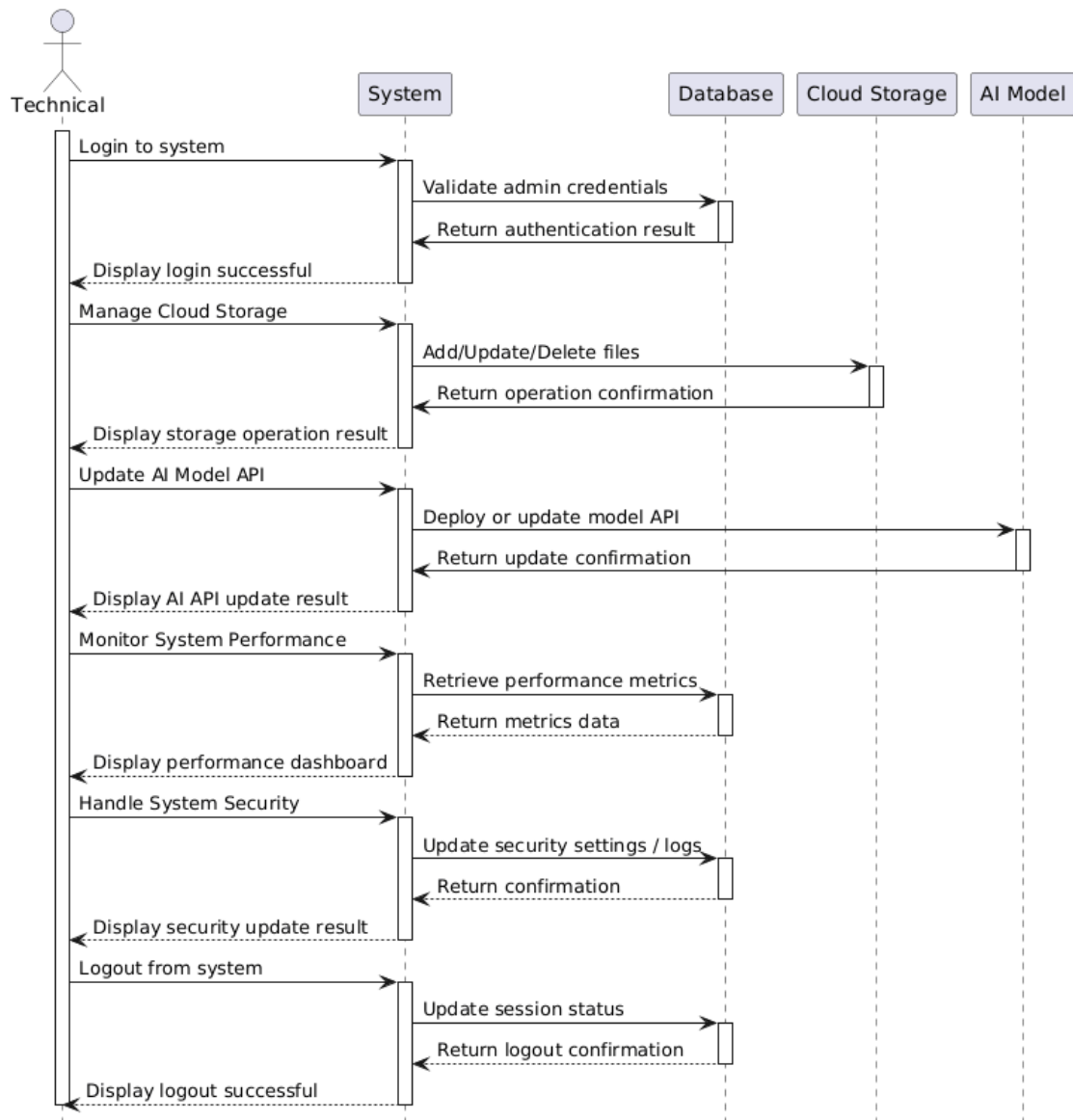


Figure 4-6: Technical Sequence Diagram

4.5.3 Activity Diagram

Activity diagrams will be used to model the system's operational workflows. They provide a visual representation of the step-by-step activities and decision points that occur during key processes, as illustrated in the project's activity diagram (Figure 4-7).

Admin - Activity Diagram

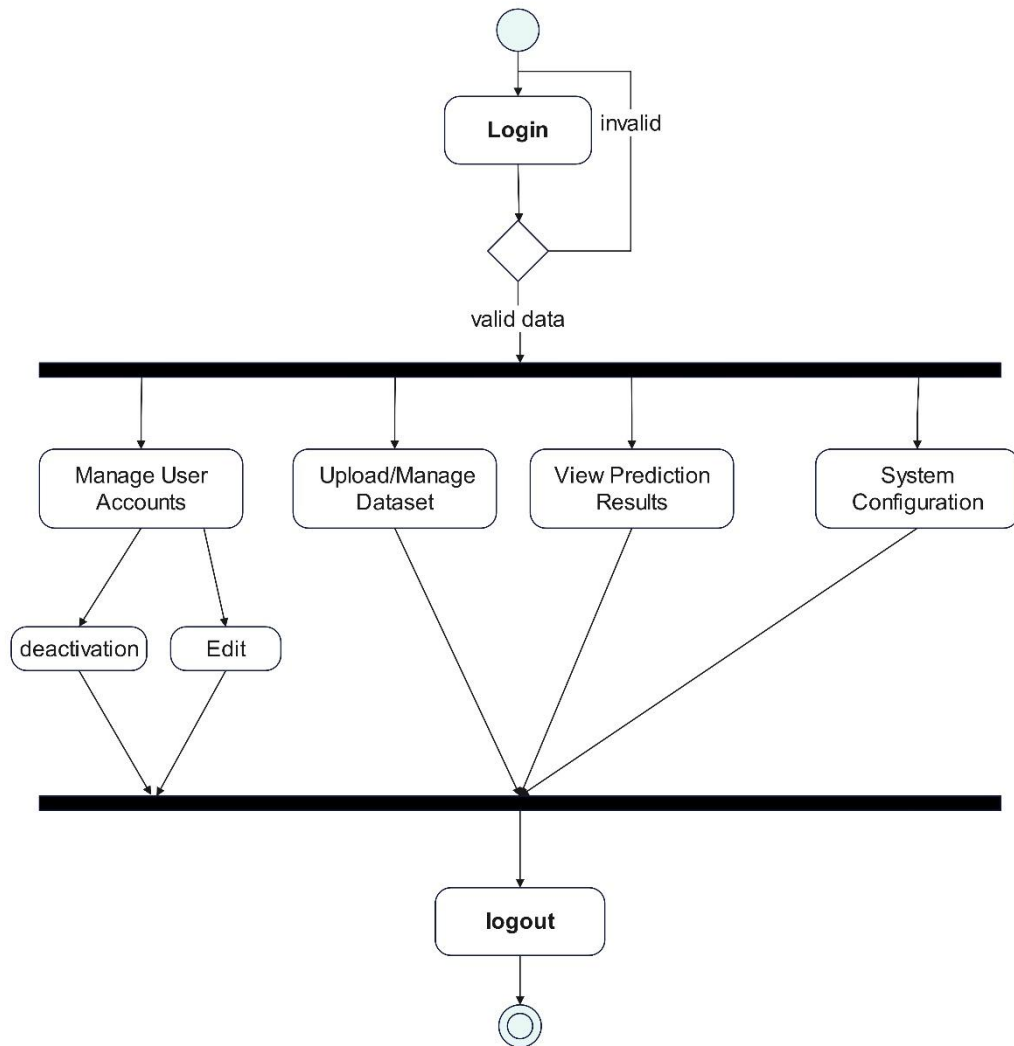


Figure 4-7: Admin Activity Diagram

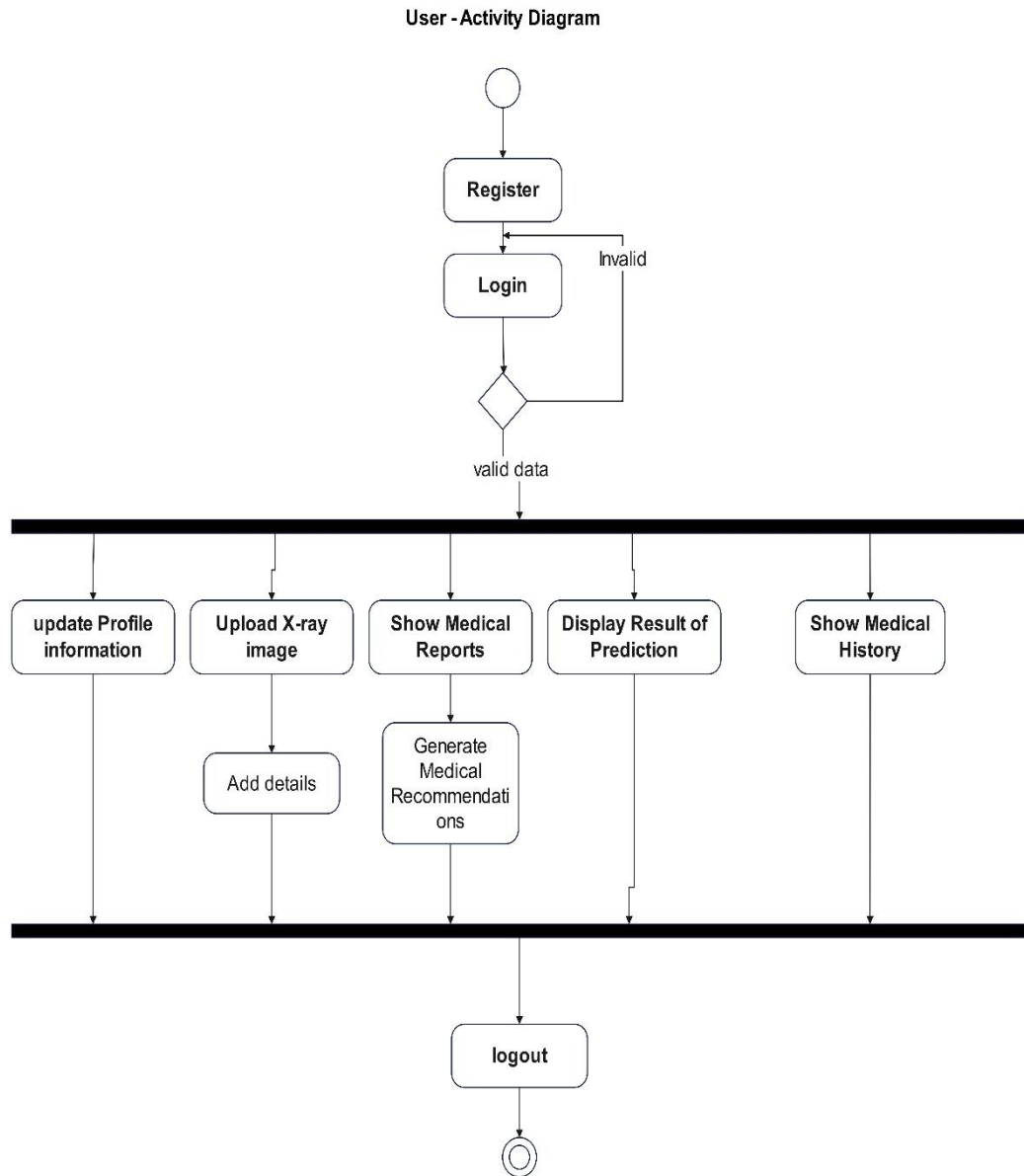


Figure 4-8: User Activity Diagram

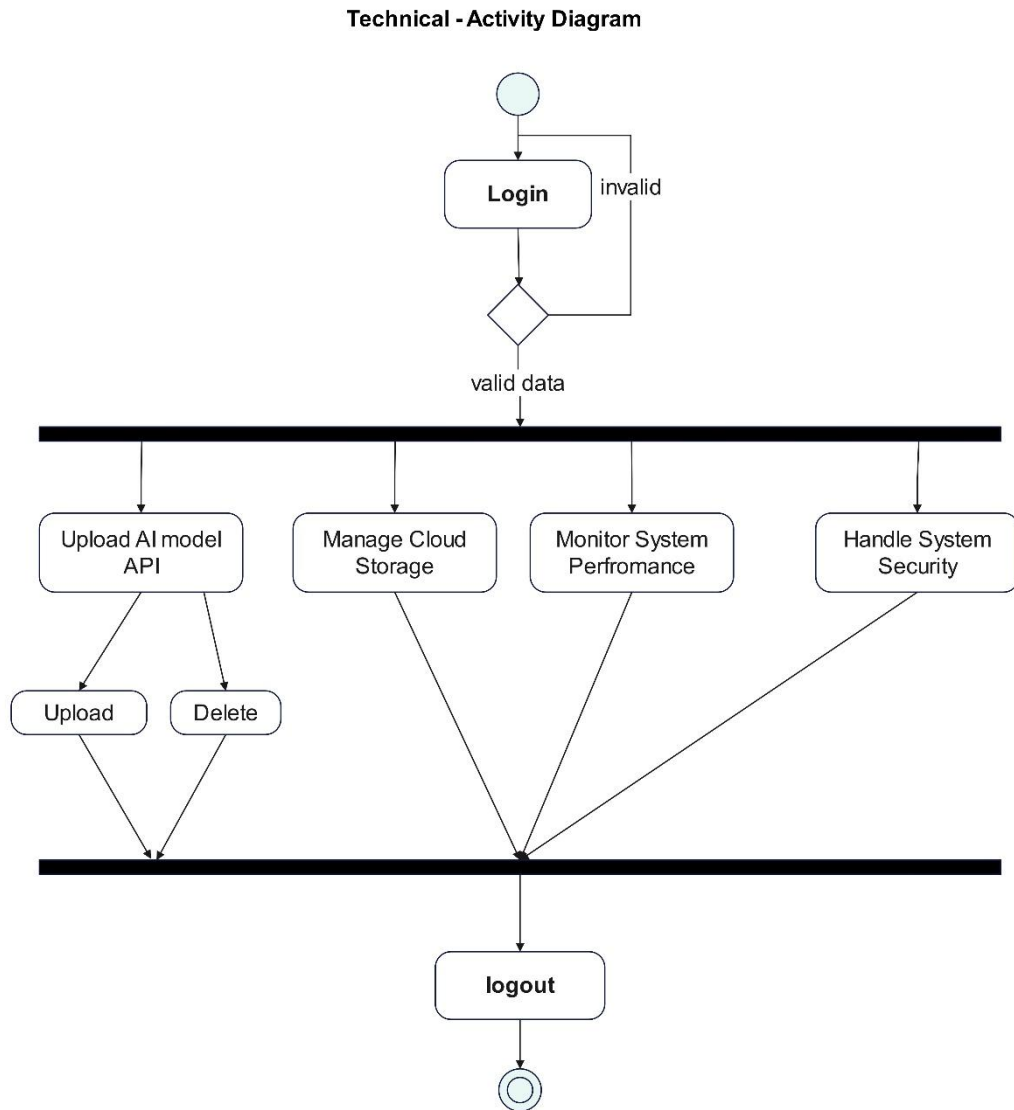


Figure 4-9: Technical Activity Diagram

4.6 Conclusion

This chapter detailed the system's requirements, and essential app components including functional and non-functional specifications, hardware and software requirements for developing application, and UML diagrams.

CHAPTER 5: SYSTEM DESIGN

5.1 Introduction

In this chapter, we present the overall system design of our project, outlining the architectural structure, system components, product features, and data storage mechanisms. The purpose of this design is to translate the system requirements into a clear blueprint that guides implementation. It includes user interface structure, database schema, high-level diagrams, and class relationships.

5.2 Product Features

In our project provides a set of features designed to support early detection of pneumonia using artificial intelligence. The main features of the system include:

- **AI-Based Pneumonia Detection:** Utilizes a trained deep learning model to analyze chest X-ray images, classify the presence of pneumonia, and generate a confidence score.
- **Chest X-ray Upload Module:** Allows users to upload medical images in standard formats (JPG, PNG), which are processed and evaluated by AI model.
- **Risk Rating & Diagnostic Insights:** Generates a risk score indicating the severity level (Low, Medium, High) and provides medically oriented insights to assist decision-making.
- **Secure User Account Management:** Supports user and admin authentication, including registration, login, logout, and password validation.
- **Medical History Tracking:** Stores previous prediction results, uploaded images, and timestamps, enabling users to view their diagnostic timeline.
- **Admin Dashboard & Dataset Management:** Provides administrators with tools to manage user accounts, update datasets, retrain AI models, and monitor system performance.

- **Data Privacy & Security Mechanisms:** Ensures sensitive patient data is encrypted and protected from unauthorized access, aligned with healthcare security standards.

5.3 USER INTERFACE

The system's user interface is designed to be simple, intuitive, and medically appropriate for users. It consists of:

1. Welcome Interface:

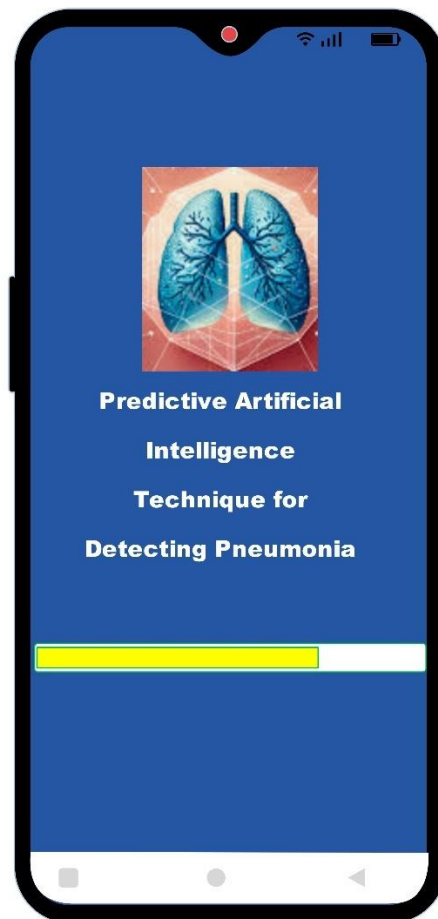
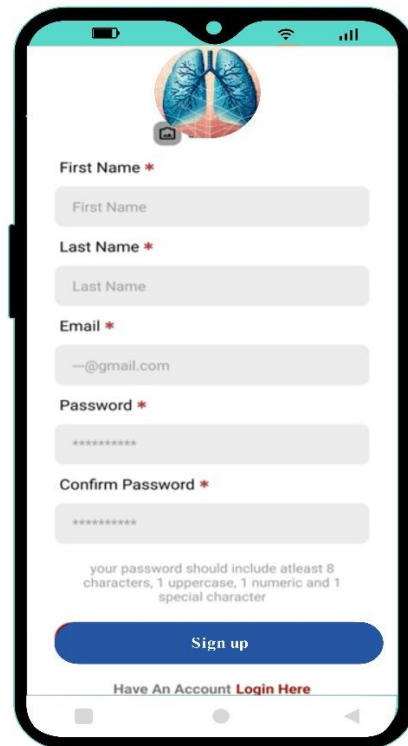


Figure 5-1: Welcome Interface

2. Register Interface:



A mobile app mockup for a registration screen. At the top is a header with a lung illustration and a camera icon. Below are input fields for First Name, Last Name, Email, Password, and Confirm Password, each with a red asterisk indicating a required field. The email field has a placeholder '@gmail.com'. The password fields have placeholder asterisks. A note below the password fields states: 'your password should include atleast 8 characters, 1 uppercase, 1 numeric and 1 special character'. At the bottom is a blue 'Sign up' button and a link 'Have An Account Login Here'.

First Name *

Last Name *

Email *

Password *

Confirm Password *

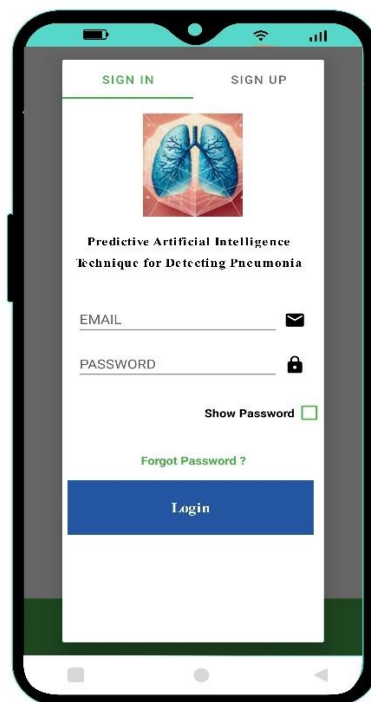
your password should include atleast 8 characters, 1 uppercase, 1 numeric and 1 special character

Sign up

Have An Account [Login Here](#)

Figure 5-2: Register Interface

3. Login Interface:



A mobile app mockup for a login screen. At the top are two tabs: 'SIGN IN' (active) and 'SIGN UP'. Below is a header with a lung illustration and the title 'Predictive Artificial Intelligence Technique for Detecting Pneumonia'. There are input fields for EMAIL and PASSWORD, each with an icon (envelope and lock respectively). A 'Show Password' checkbox is next to the password field. Below is a link 'Forgot Password ?' and a blue 'Login' button.

SIGN IN SIGN UP

Predictive Artificial Intelligence Technique for Detecting Pneumonia

EMAIL

PASSWORD

Show Password ☐

[Forgot Password ?](#)

Login

Figure 5-3: Login Interface

4. Dashboard Interface:

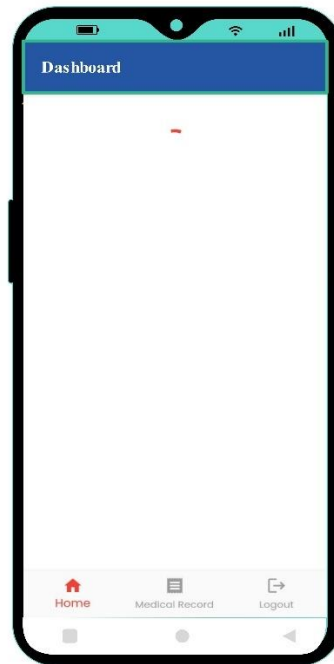


Figure 5-4: Dashboard Interface

5. Medical History Interface:



Figure 5-5: Medical History Interface

5.4 Data Storage

In our project, the data storage design follows a structured relational database model to ensure secure, organized, and efficient management of user information, X-ray uploads, prediction outputs, and historical records. The database consists of 6 main tables: Admin, User, Health Data, Prediction, and Medical History.

5.4.1 Admin Table

Stores administrator accounts responsible for overseeing the system.

Table 5-1: Admin Table

Field	Type	Note
Admin_ID	INT (10)	Primary Key
Email	VARCHAR(50)	Not Null
Password	VARCHAR(50)	NOT NULL

5.4.2 User Table

Contains information for registered patients using the system.

Table 5-2: User Table

Field	Type	Note
User_ID	INT (10)	Primary Key
Full_Name	VARCHAR(50)	Not Null
Email	VARCHAR(50)	Unique, Not Null
Password	VARCHAR(50)	NOT NULL

Phone	VARCHAR(20)	NOT NULL
Date_Joined	DATETIME	NOT NULL

5.4.3 Health Data Table

Stores information related to uploaded chest X-ray images.

Table 5-3: Health Data Table

Field	Type	Note
Image_ID	INT (10)	Primary Key
User_ID	INT (10)	Foreign Key → User Table
Image_Path	VARCHAR(255)	NOT NULL
Upload_Date	DATETIME	NOT NULL
Image_Format	VARCHAR(10)	NOT NULL

5.4.4 Prediction Table

Contains prediction and AI model output for each uploaded X-ray.

Table 5-4: Prediction Table

Field	Type	Note
Prediction_ID	INT (10)	Primary Key
Image_ID	INT (10)	Foreign Key → Health Data
Result	VARCHAR(20)	NOT NULL
Confidence_Score	FLOAT	NOT NULL
Risk_Level	VARCHAR(20)	NOT NULL
Prediction_Date	DATETIME	NOT NULL

5.4.5 Medical History Table

Tracks chronological prediction results for each user.

Table 5-5: Medical History Table

Field	Type	Note
History_ID	INT (10)	Primary Key
User_ID	INT (10)	Foreign Key
Prediction_ID	INT (10)	Foreign Key
Notes	TEXT	NOT NULL
Record_Date	DATETIME	NOT NULL

5.4.6 Smart Health Tips Table

Table 5-6: Smart Health Tips Table

Field	Type	Description
Tip_ID	INT (10)	Primary Key
Tip_Text	TEXT	NOT NULL
Category	VARCHAR(30)	NOT NULL
Risk_Level	VARCHAR(20)	NOT NULL
Created_Date	DATETIME	NOT NULL
Admin_ID	INT (10)	Foreign Key

5.5 High Level Design

5.5.1 ER Diagram

The Entity-Relationship (ER) Diagram illustrates how system entities such as Users, Admin, X-ray Images, and Prediction Results are connected. As shown in Figure 5-6. It shows relationships between users and their medical history, and relationships between uploaded images and generated predictions.

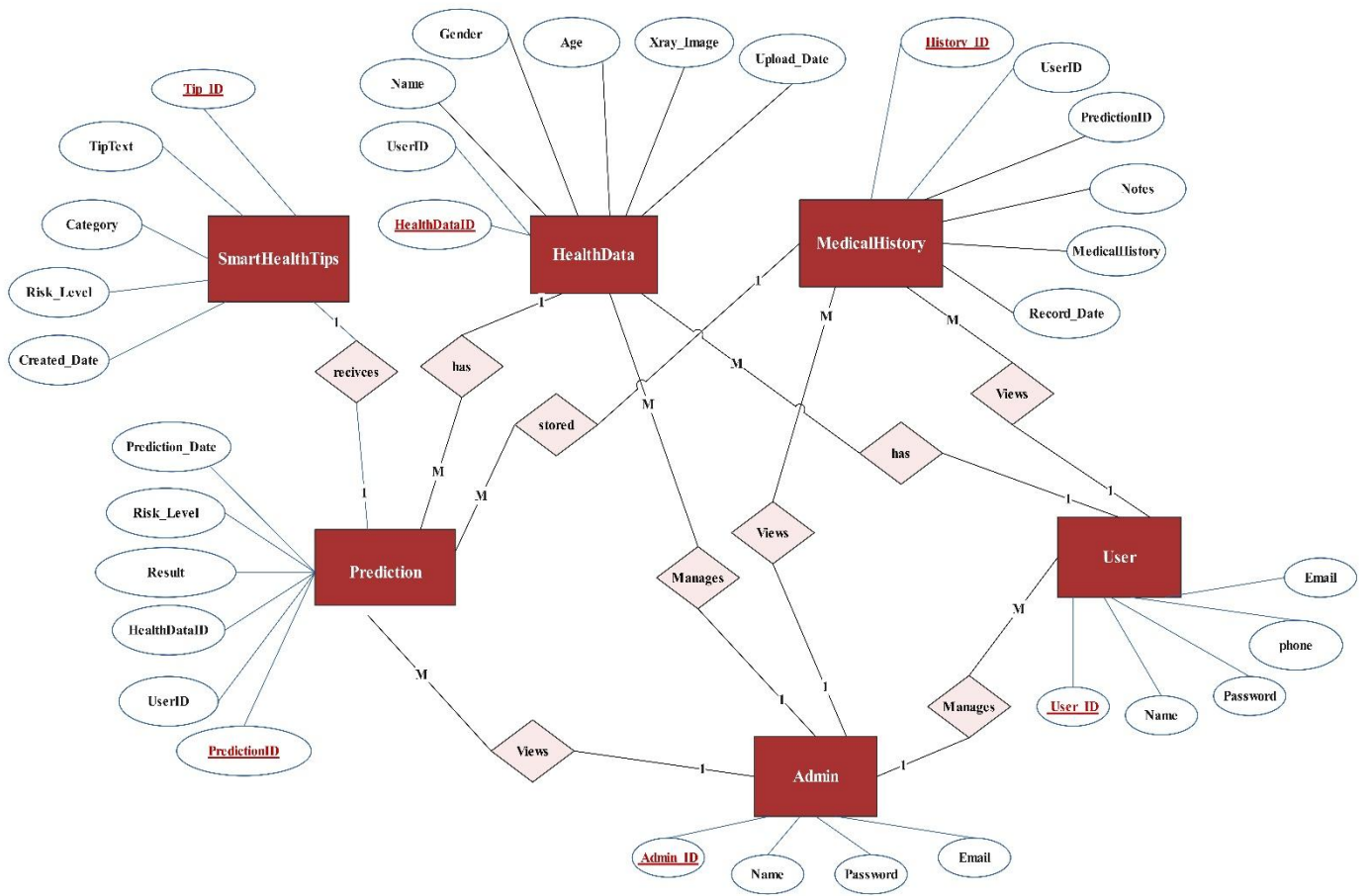


Figure 5-6: ER Diagram

5.5.2 Class Diagrams

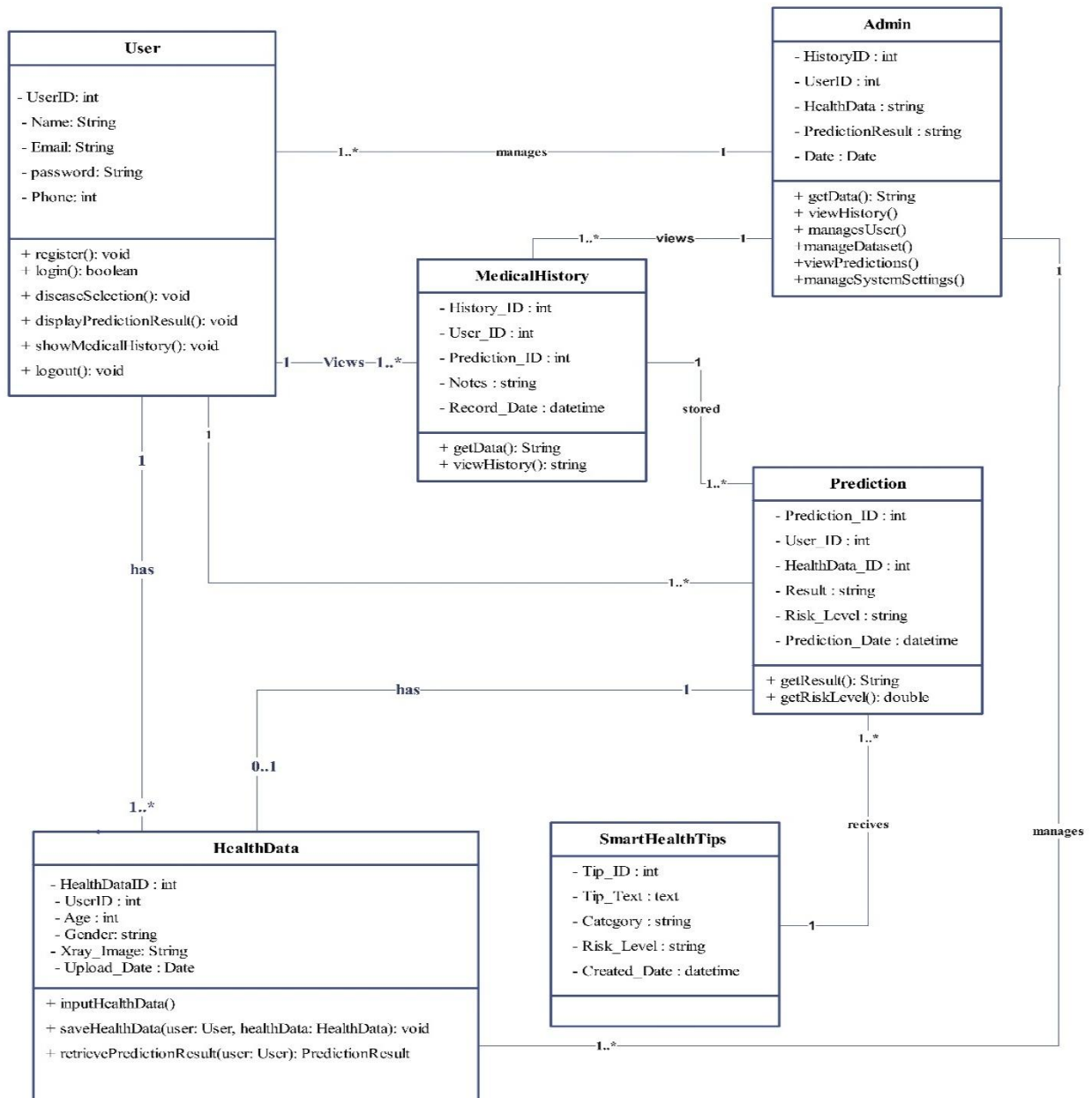


Figure 5-7: Class Diagram

5.6 Conclusion

This chapter defined the system design framework for our project, covering its features, interface structure, data storage model, and high-level architecture. The diagrams and descriptions presented here transform the requirements into a practical implementation plan.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Future Work

- The next phase will focus on developing the complete system interfaces using Android Studio for both user and admin interaction.
- A mobile-friendly application will be created to ensure smooth and accessible usage.
- The AI model will be integrated into the mobile app to provide accurate pneumonia prediction from uploaded chest X-ray images.
- In the upcoming semester, the full system will progress through all remaining project stages, including implementation and system testing.

6.2 Summary

In conclusion, our project aims to provide an intelligent and accessible solution for early detection of pneumonia using deep learning techniques. Throughout the development process, we analyzed existing systems, defined requirements, designed system components, and developed a structured plan to implement an accurate and user-friendly diagnostic application. With the planned development, testing, and refinement in the next semester.

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